Tor Vergata University of Rome

School of Economics

Department of Economics and Finance

From Mobility Data to Investment Governance

A Risk-Adjusted Financial and Economic Framework for Smart Urban Infrastructure in Euro-Mediterranean Cities

Research Proposal for the M.Sc. Finance and Banking

Applicant: El Ghali Sany

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Administrative Information

Table 1. Candidate and Proposal Metadata

Field	Details
Applicant	Elghali Sany
Email	sanyghali@gmail.com
Academic qualifications	Bachelor's in Management, Finance and Accounting; MSc candidate in Procurement and Supply Chain Management
Department	Department of Economics and Finance
School	School of Economics
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Abstract

Smart urban infrastructure investment increasingly depends on the capacity to convert mobility data into credible financial and governance decisions. Yet municipal and private-sector evaluations often remain fragmented: traffic engineers model mobility flows, financial analysts calculate profitability, regulators assess compliance, and public-private partnership actors negotiate risk through separate analytical systems. This research proposal develops a risk-adjusted investment evaluation framework for smart urban infrastructure in Euro-Mediterranean cities, using Casablanca as the empirical anchor and EU/Italian governance conditions as a regulatory benchmark. The proposed framework integrates mobility-derived exposure indicators, CAPEX/OPEX-based financial modelling, multi-criteria decision analysis, and regulatory-governance scoring into a composite Investment Attractiveness Score. Methodologically, the study combines GIS/SUMO-supported spatial analysis, discounted cash-flow metrics, AHP-based weighting, TOPSIS/COPRAS ranking logic, and sensitivity analysis across traffic, revenue, cost, discount-rate, data-quality, and regulatory scenarios. The expected outputs are a location ranking matrix, a profitability-risk quadrant, a governance-adjusted investment score, and policy recommendations for municipalities, investors, PPP operators, and urban planners. The proposal contributes to applied urban economics and infrastructure finance by linking spatial demand estimation with financial viability and regulatory risk, while remaining bounded to mobility-dependent urban assets rather than all smart-city technologies.

Keywords: smart urban infrastructure; mobility data; infrastructure finance; public-private partnerships; MCDA; AHP; TOPSIS; Euro-Mediterranean governance; CAPEX/OPEX analysis; Investment Attractiveness Score.

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1. Introduction

Cities are under pressure to invest in smarter, more data-driven infrastructure, but the tools used to evaluate these investments have not kept pace with the complexity of the decision. A municipality or private operator deciding where to deploy digital advertising panels, smart transit shelters, connected street furniture, or mobility-linked urban assets must answer four questions simultaneously: is the location exposed to enough traffic or pedestrian flow; can this exposure generate credible revenue; do expected cash flows justify capital and operating costs; and can the project operate within procurement, data-protection, and public-private partnership rules?

Current evaluation practice rarely answers these questions within a single analytical framework. Traffic engineers may estimate flows. Financial analysts may calculate net present value, return on investment, and payback period. Regulators may assess procurement and data compliance. Urban planners may prioritise locations according to accessibility or public-service value. Each analysis is useful, but each remains incomplete when used alone. The result is a fragmented decision process in which mobility data, financial viability, regulatory exposure, and governance quality are treated as separate problems, although investors and municipalities experience them as one integrated risk.

This fragmentation matters because smart urban infrastructure is not merely a technological category. In this proposal, smart urban infrastructure refers to physical urban assets whose economic value depends partly on digital systems, mobility exposure, or data-enabled operation. Examples include digital out-of-home advertising panels, connected bus shelters, smart mobility nodes, and sensor-supported street furniture. These assets require capital expenditure, meaning the initial cost of equipment, installation, permits, and deployment, and they generate operating expenditure, meaning recurring costs such as maintenance, energy, connectivity, data management, and supervision. Their financial viability depends on whether future cash flows can justify these costs over time.

Mobility data is therefore not a secondary technical input. It is a direct investment variable. Traffic intensity, pedestrian movement, dwell time, line-of-sight exposure, public transport proximity, and spatial accessibility shape the demand potential of mobility-dependent infrastructure. For advertising-linked urban assets, a high-traffic location can increase exposure and revenue potential. However, traffic alone does not make a site investable. A location may appear attractive because it is visible, yet become financially weak once installation costs, concession fees, regulatory restrictions, data limitations, or governance delays are included in the evaluation.

Previous research and policy work provide important foundations for this problem. Urban mobility frameworks increasingly emphasise standardised mobility data and sustainable transport indicators. Infrastructure finance literature provides established tools such as NPV, ROI, payback period, and sensitivity analysis. Public-private partnership research examines risk allocation, procurement transparency, and value-for-money assessment. Data governance literature highlights privacy, data access, compliance, and institutional accountability. Multi-Criteria Decision Analysis methods, including AHP, TOPSIS, and COPRAS, provide techniques for ranking alternatives under multiple criteria. Yet these bodies of knowledge remain insufficiently connected in the specific context of smart urban infrastructure investment.

The central gap is therefore not the absence of methods. The gap is the absence of integration. Existing approaches often examine smart infrastructure, mobility data, financial profitability, regulatory constraints, and governance risk through separate disciplinary lenses. The financial evaluation of urban infrastructure assets rarely treats mobility-derived demand as a core revenue driver. Mobility-based exposure models rarely translate directly into CAPEX/OPEX-based investment evaluation. Regulatory and data-governance constraints are often treated as compliance issues rather than variables that alter financial feasibility. MCDA models are frequently applied to infrastructure prioritisation, but financial risk and regulatory friction are often under-specified as scoring dimensions.

This research addresses that gap by developing a risk-adjusted investment evaluation framework for smart urban infrastructure in Euro-Mediterranean cities. The framework integrates five dimensions: Mobility Potential, Financial Viability, Risk Adjustment, Governance Quality, and Data Quality. These dimensions are combined into an Investment Attractiveness Score, a composite indicator designed to compare locations or project alternatives according to their economic, spatial, financial, regulatory, and institutional conditions.

The empirical anchor of the study is Casablanca, where the researcher has prior applied experience in evaluating advertising infrastructure profitability using road traffic data, GIS spatial analysis, SUMO traffic simulation, CAPEX/OPEX modelling, and regulatory review. This case provides a practical basis for developing the framework. The European Union, and particularly Italy as a more codified procurement and regulatory environment, provides a comparative benchmark for assessing how governance maturity and data regulation affect investment attractiveness. This comparative logic supports the proposal's Euro-Mediterranean focus, where infrastructure connectivity, private investment mobilisation, and regulatory divergence remain central policy issues.

The study is deliberately bounded. It does not attempt to evaluate all smart city technologies, nor does it claim to solve the full infrastructure finance gap across the Euro-Mediterranean region. It focuses on mobility-dependent urban infrastructure assets whose viability can be evaluated through spatial exposure, financial modelling, governance assessment, and risk-adjusted scoring. This boundary makes the project empirically feasible and analytically precise.

The main research question is: How can mobility-derived demand indicators, financial viability metrics, and regulatory-governance risk variables be integrated into a risk-adjusted evaluation framework for mobility-dependent smart urban infrastructure investment, using Casablanca as the empirical anchor and EU/Italian regulatory conditions as a comparative benchmark?

To answer this question, the study pursues three objectives. First, it develops a structured method for converting mobility and spatial exposure data into investment-relevant demand indicators. Second, it links these indicators to financial evaluation metrics, including CAPEX, OPEX, NPV, ROI, IRR, and payback period. Third, it incorporates regulatory risk, governance quality, and data reliability into a multi-criteria scoring model capable of ranking investment alternatives.

The expected contribution is both academic and practical. Academically, the study connects urban mobility analytics, infrastructure finance, data governance, and public-private partnership research within a single investment evaluation model. Practically, it provides municipalities, private investors, PPP operators, and urban planners with a decision-support framework for identifying projects that are not only visible or technologically attractive, but financially viable, legally feasible, and institutionally robust.

The integrated evaluation logic is developed visually in the conceptual framework and analytical workflow figures below; no standalone visual is inserted at this stage to avoid interrupting the introduction.

2. Background and Rationale

2.1 The investment problem behind smart urban infrastructure

Smart urban infrastructure refers to physical urban assets whose operation or value depends on digital systems, mobility data, or connected services. This includes digital out-of-home advertising panels, smart bus shelters, connected street furniture, urban mobility nodes, data-enabled transport infrastructure, and sensor-based municipal assets. These projects are not only technological upgrades. They are capital-intensive investments that require credible demand estimation, financial viability, regulatory compliance, and public-private coordination.

This distinction is critical. A city can deploy a smart asset because it appears modern, but an investor or municipality must evaluate whether the asset can generate enough value over time to justify its capital expenditure. CAPEX refers to the initial investment required to acquire and install the infrastructure, while OPEX refers to recurring operating costs such as maintenance, electricity, connectivity, software, and supervision. NPV and ROI measure whether expected future cash flows justify the investment. However, these financial indicators become unreliable if the underlying demand assumptions are weak.

In smart urban infrastructure, demand is increasingly measured through mobility data: traffic intensity, pedestrian flow, dwell time, public transport proximity, line-of-sight exposure, and spatial accessibility. For assets such as digital advertising panels or connected mobility shelters, traffic is not merely a transport variable; it becomes an economic input. It conditions exposure, revenue potential, concession value, and investment risk.

2.2 Why the issue is urgent now

The timing of this research is not arbitrary. Urban infrastructure investment is entering a period in which fiscal pressure, digital transformation, and regulatory complexity are converging. The European Investment Bank Municipalities Survey 2022-2023 reported that 90% of surveyed EU municipalities planned to increase infrastructure investment in at least one area over the following three years, while investment adequacy remained a concern across multiple infrastructure domains (European Investment Bank, 2023). This indicates that cities are not only discussing infrastructure renewal; they are actively preparing investment decisions under constrained financial and administrative conditions.

At the same time, smart city implementation has exposed a governance failure. The European Court of Auditors concluded in Special Report 24/2023 that EU smart city initiatives produced tangible solutions, but their wider adoption was limited by fragmentation, coordination weaknesses, and insufficient quantitative indicators for measuring impact (European Court of Auditors, 2023). The implication for this research is direct: if smart city projects lack robust quantitative decision frameworks, municipalities and investors cannot reliably compare alternative projects, locations, or governance models.

Data governance adds a second layer of urgency. The OECD identifies insufficient financial resources, weak business models for financing data collection, limited compliance with data-protection legislation, and security risks as core barriers to smart city data governance (OECD, 2023). This matters because mobility data is not a neutral technical input. It is a regulated asset. Its availability, reliability, legal status, and processing constraints directly affect the credibility of revenue forecasts for data-driven infrastructure.

The Euro-Mediterranean policy window strengthens the "why now" argument. The European Commission's Global Gateway strategy is designed to mobilise up to EUR 300 billion in infrastructure-related investments between 2021 and 2027, including digital and transport connectivity (European Commission, 2021a; European Commission, 2026). The OECD and Union for the Mediterranean also identify infrastructure connectivity and governance improvement as priorities for regional integration (OECD & Union for the Mediterranean, 2025). Therefore, this research is urgent because investment decisions are being made now, while the tools used to appraise smart urban infrastructure remain fragmented.

2.3 Casablanca as an emerging urban laboratory

Casablanca is a relevant empirical anchor because it combines dense mobility demand, urban transformation, digitalisation, and evolving public-private investment needs. It should not be presented vaguely as "interesting" or "fast-growing"; it must be framed as a live urban laboratory where mobility-based infrastructure valuation is practically necessary. The Casablanca-Settat region is Morocco's most populated region according to the 2024 General Population and Housing Census results published by the High Commission for Planning (Haut-Commissariat au Plan, 2024).

Casablanca has also moved toward urban data infrastructure. Minsait, an Indra company, announced that it had won a contract to develop Casablanca's urban data management platform with Casablanca Prestations, a local development company of Casablanca City Council. The platform was presented as one of the pillar projects of the city's digital transformation and as a benchmark for urban modernisation in Morocco (Minsait, 2019). This supports the claim that Casablanca is not merely a theoretical case; it has an existing digital governance trajectory.

Casablanca is also analytically valuable because the researcher has prior applied experience there. The previous project evaluated advertising infrastructure profitability using road traffic data, GIS spatial analysis, SUMO traffic simulation, CAPEX/OPEX modelling, and regulatory analysis. This gives the proposed research a credible empirical foundation: it extends an existing applied case into a more general investment governance framework.

2.4 The regulatory comparison: why Italy and the EU matter

Italy and the wider EU provide a useful regulatory benchmark because they represent a more codified environment for public procurement, data governance, and urban mobility planning. The EU Urban Mobility Framework encourages structured mobility planning and sustainable urban mobility indicators, including affordability of public transport, road safety, emissions, congestion, and modal split (European Commission, 2021b). These indicators show that mobility data is becoming part of formal urban governance, not merely an optional analytical input.

Italy adds a concrete procurement reference point. Legislative Decree No. 36/2023 established the new Italian Public Contracts Code, covering procurement and concession contracts. For infrastructure concessions, such frameworks matter because procurement rules define risk transfer, contractual autonomy, public accountability, and the legal conditions under which private operators can invest (Italian Republic, 2023).

Morocco's PPP framework is also evolving. The World Bank's PPP legal database states that Law No. 46-18 modified and supplemented Law No. 86-12 on public-private partnership contracts (World Bank, 2024a). This makes the Morocco-EU comparison legitimate: both contexts are institutionally active, but they operate with different levels of codification, data governance maturity, and procurement predictability.

The point of comparison is not to claim that Europe is universally "better" and Morocco is "weaker". The point is to measure how different governance environments affect the same investment question: when does a smart urban infrastructure project become financially viable, legally executable, and institutionally attractive?

2.5 From context to research necessity

The rationale for this research is not that smart cities are fashionable. The rationale is that cities increasingly need to decide where and how to deploy smart urban infrastructure, while the decision tools used to evaluate these investments remain fragmented.

Traffic engineers can estimate mobility flows. Financial analysts can calculate NPV and ROI. Regulators can assess compliance. PPP lawyers can review concession structures. But these analyses often remain disconnected. A high-traffic site may appear profitable but become unattractive once permitting delays, weak data quality, privacy compliance costs, or demand volatility are included. Conversely, a moderate-traffic site may be more investable if governance quality is higher, data are reliable, and operating costs are lower.

This research creates value for four groups. Municipal authorities gain a transparent tool for prioritising smart infrastructure projects and evaluating PPP proposals. Private investors and infrastructure operators gain a risk-adjusted screening model before committing to costly due diligence. Urban planners gain a method for linking mobility data to infrastructure siting decisions. Researchers in applied economics, infrastructure finance, and urban governance gain a replicable framework connecting demand estimation, financial modelling, and regulatory risk.

3. Research Problem and Research Questions

3.1 Problem Statement

Smart urban infrastructure investment decisions increasingly depend on the capacity to connect spatial demand, financial viability, and governance risk within a single evaluation logic. However, current investment assessment practices remain analytically fragmented. Mobility data are often used to estimate traffic flows, pedestrian exposure, or location attractiveness, but they are rarely translated into project-level financial indicators such as CAPEX, OPEX, NPV, ROI, and payback period. Conversely, financial feasibility studies often calculate investment returns without systematically incorporating data quality, demand volatility, regulatory constraints, or public-private partnership governance conditions.

This creates a research problem rather than merely a managerial inconvenience. The unresolved issue is not simply that municipalities or investors lack a ready-made decision tool. The deeper problem is that existing evaluation approaches do not clearly explain how mobility-derived demand indicators, financial viability metrics, regulatory constraints, and governance-risk variables can be jointly operationalised within one coherent investment framework. As a result, a high-traffic location may be incorrectly interpreted as financially attractive, while a seemingly profitable project may become unviable once data restrictions, procurement uncertainty, compliance costs, or governance weaknesses are incorporated.

This study addresses that problem by developing and testing a bounded risk-adjusted evaluation framework for mobility-dependent smart urban infrastructure. The empirical focus is limited to Casablanca as the primary urban anchor, with EU/Italian regulatory conditions used as a comparative benchmark. The study focuses on infrastructure assets whose investment value depends directly on spatial exposure and mobility patterns, such as digital out-of-home advertising panels, smart bus shelters, connected street furniture, and mobility-linked urban nodes. It does not attempt to evaluate all smart city technologies or all Euro-Mediterranean infrastructure systems. Instead, it examines how a specific class of mobility-dependent urban assets can be assessed through an integrated model combining mobility potential, financial viability, risk adjustment, governance quality, and data quality.

The research therefore responds directly to the gap identified in the literature: analytical tools exist, but they are usually applied separately. Smart city research often lacks financial operationalisation; infrastructure finance often lacks mobility-data integration; MCDA applications often under-specify financial and governance risk; and data governance studies rarely connect legal constraints to investment evaluation.

3.2 Main Research Question

> **How can mobility-derived demand indicators, financial viability metrics, and regulatory-governance risk variables be integrated into a risk-adjusted evaluation framework for mobility-dependent smart urban infrastructure investment, using Casablanca as the empirical anchor and EU/Italian regulatory conditions as a comparative benchmark?**

3.3 Specific Research Sub-questions

- SQ1 - Mobility-to-demand conversion: How can urban mobility indicators, including traffic intensity, pedestrian flow, spatial exposure, dwell time, accessibility, and public transport proximity, be converted into investment-relevant demand indicators for mobility-dependent smart urban infrastructure?
- SQ2 - Financial viability measurement: Which financial indicators - CAPEX, OPEX, NPV, ROI, IRR, and payback period - are most appropriate for evaluating the viability of selected smart urban infrastructure assets under different demand and cost assumptions?
- SQ3 - Risk and governance operationalisation: How can demand uncertainty, data quality, regulatory constraints, and PPP governance conditions be translated into measurable risk-adjustment variables within the investment evaluation model?
- SQ4 - Composite scoring and decision output: To what extent can a multi-criteria Investment Attractiveness Score rank selected infrastructure locations or project alternatives in a transparent, replicable, and sensitivity-tested manner?

3.4 Ring-fencing and Feasibility Boundaries

The study is ring-fenced in four ways. Geographically, it uses Casablanca as the empirical anchor and EU/Italian regulatory conditions as the benchmark rather than attempting to cover all Euro-Mediterranean cities. Conceptually, it focuses on mobility-dependent smart urban assets rather than the full smart city technology universe. Methodologically, it develops an evaluation framework based on available mobility, financial, governance, and regulatory indicators rather than claiming deterministic prediction of actual future revenues. Temporally, it treats the project as a framework development and pilot-testing study, with empirical calibration dependent on the availability of site-level mobility, cost, revenue, and regulatory data.

4. Literature Review

This literature review is organised thematically rather than chronologically. Its purpose is not to summarise isolated authors, but to examine how six research streams jointly frame the proposed study: smart city infrastructure, mobility-data analytics, infrastructure finance, multi-criteria decision analysis, PPP governance, and data governance. The review argues that these streams are complementary but still insufficiently integrated for evaluating mobility-dependent smart urban infrastructure investment.

4.1 From smart city infrastructure to investment assets

The smart city literature has expanded rapidly around digital platforms, sensors, Internet of Things devices, data hubs, mobility platforms, and citizen-facing services. This literature is valuable because it explains how urban data can support better service delivery, sustainability, and governance. UN-Habitat's World Smart Cities Outlook 2024, for instance, frames smart city development around people-centred digital transformation, urban governance, and digital inclusion rather than around technology alone (UN-Habitat, 2024).

However, this literature often treats smart infrastructure as a public innovation or digital-governance problem rather than as an investment asset. This is a major limitation for the present research. Smart shelters, digital out-of-home advertising panels, connected street furniture, and mobility-linked urban nodes are not only technological artefacts. They require capital expenditure, operating expenditure, concession arrangements, maintenance contracts, data systems, and revenue assumptions. A smart asset may be technologically desirable but financially weak if it cannot produce credible cash flows or if the institutional conditions around it are unstable.

The contribution of this research begins at this tension. Instead of asking whether smart infrastructure is desirable in general, the study asks how its investment attractiveness can be assessed when financial viability depends simultaneously on spatial exposure, demand conversion, data quality, regulatory constraints, and governance risk.

4.2 Mobility data as a demand-estimation tool

A second research stream examines mobility data as a way to estimate urban demand. Traffic counts, GPS traces, mobile phone location data, smart-card records, bus GPS data, land-use data, and GIS-based spatial indicators help researchers identify where people move, when flows intensify, and how accessibility or visibility changes across urban space. This literature provides the demand-estimation foundation for the proposed framework.

Recent empirical work on targeted transit and outdoor advertising demonstrates the relevance of these methods. Huang et al. (2022) used mobile phone location data, land-use data, bus network data, bus GPS data, and smart-card data to identify bus routes likely to maximise advertising effectiveness.

This type of study is directly relevant because it shows that mobility datasets can estimate audience exposure and location attractiveness in advertising-linked infrastructure contexts.

Yet the limitation is clear: most mobility-based advertising and exposure studies stop at audience or exposure optimisation. They can answer where the audience is likely to be, but they rarely answer whether the expected exposure justifies CAPEX, OPEX, concession fees, financing costs, regulatory compliance, and long-term operational risk. In investment terms, exposure is only an upstream input. It becomes useful for decision-making only when translated into revenue assumptions and tested against financial and governance constraints.

4.3 Infrastructure finance and the limits of conventional appraisal

Infrastructure finance provides the valuation tools needed to evaluate capital-intensive assets. Net Present Value, Return on Investment, Internal Rate of Return, payback period, CAPEX/OPEX analysis, and sensitivity testing remain essential for assessing whether expected future cash flows justify an initial investment. Kumar, Srivastava, and Tabash's systematic literature review of infrastructure project finance identifies research domains around capital markets, investment performance, policy and stakeholder perspectives, and institutional frameworks, while calling for more holistic understanding of infrastructure project finance under uncertainty (Kumar et al., 2021).

The weakness is not that infrastructure finance lacks rigour. Its weakness, for this specific proposal, is scope. The literature remains more mature for large-scale infrastructure such as roads, ports, energy systems, airports, and utilities than for smaller-scale, mixed-revenue smart urban assets. Digital panels, smart shelters, and connected street furniture often depend on advertising demand, public-space concessions, mobility exposure, data availability, and local regulatory permissions. These variables are more granular and more volatile than the demand assumptions used in many traditional infrastructure assets.

Consequently, conventional appraisal is necessary but insufficient. NPV and ROI can measure projected profitability, but they cannot independently verify whether the demand assumptions behind projected revenue are reliable. Nor do they automatically incorporate data governance, procurement quality, PPP transparency, or regulatory restrictions. This research therefore embeds conventional financial indicators inside a wider risk-adjusted framework rather than replacing them.

4.4 Multi-criteria decision analysis: useful ranking logic, incomplete financial integration

Multi-Criteria Decision Analysis (MCDA) offers a methodological bridge because infrastructure investment decisions rarely depend on one criterion. The Analytic Hierarchy Process, associated with Saaty's work, helps structure expert judgement and derive weights among competing criteria. TOPSIS, formalised in the multiple-attribute decision-making tradition by Hwang and Yoon, ranks alternatives according to closeness to an ideal and distance from a negative-ideal solution. COPRAS is also relevant because it can distinguish between benefit criteria and cost or risk criteria.

Recent GIS-MCDA applications confirm that spatial analysis and multi-criteria ranking can support urban decision-making. Their strength is the ability to combine heterogeneous criteria such as accessibility, environmental quality, cost, risk, and social impact. However, many MCDA applications in urban studies remain oriented toward site suitability, sustainability, or planning prioritisation. They often treat financial feasibility and regulatory friction as generic scores rather than as variables connected to cash-flow uncertainty, revenue sensitivity, or PPP governance.

This study therefore uses MCDA as an operational mechanism, not as a decorative method. The intended contribution is not simply to apply AHP, TOPSIS, or COPRAS to another urban planning problem. It is to test whether a multi-criteria scoring architecture can integrate five investment-relevant dimensions: Mobility Potential, Financial Viability, Risk Adjustment, Governance Quality, and Data Quality.

4.5 Public-private partnerships and governance risk

The PPP literature provides the institutional foundation of the research. Public-private partnerships are not merely financing devices; they are risk-allocation systems. The World Bank PPP Reference Guide emphasises affordability, market assessment, risk allocation, and value for money as core components of PPP evaluation (World Bank, 2024b). This matters because smart urban infrastructure often operates through concessions, municipal contracts, public-space rights, advertising rights, or public-private service arrangements.

The strength of PPP literature is its treatment of contractual risk, public accountability, and institutional coordination. Its limitation is that it often remains disconnected from granular demand estimation. PPP analysis can explain how risks should be allocated, but it rarely measures whether a specific urban location has sufficient traffic exposure to support the revenue model. Conversely, mobility analytics can measure exposure, but it rarely models concession design, public procurement risk, or contractual governance.

The European Court of Auditors' Special Report 24/2023 reinforces this concern in the smart city context. The report concluded that smart city initiatives produced tangible solutions, but that fragmentation limited wider adoption (European Court of Auditors, 2023). For this research, the implication is direct: smart city investment can fail not only because technology is weak, but because evaluation, governance, coordination, and private-finance mobilisation remain fragmented.

4.6 Data governance as an investment constraint

Data governance is the final literature stream needed to complete the framework. Smart urban infrastructure depends on data: traffic flows, mobility traces, sensor outputs, public transport usage, geolocation records, and platform-generated operational information. The OECD's Smart City Data Governance report identifies challenges related to data management, data sharing, compliance, financing, business models, privacy, and security (OECD, 2023).

This literature is often framed as a legal or administrative issue. That framing is valid but incomplete. For investment evaluation, data governance is also a valuation issue. If mobility data is incomplete, legally restricted, poorly standardised, or inaccessible, then the demand estimates used to justify investment become less reliable. If privacy rules constrain data granularity, then projected revenues for data-driven advertising infrastructure may require adjustment. If governance is weak, investors face future regulatory, operational, and reputational risks.

This is why the proposed framework treats Data Quality and Governance Quality as explicit scoring dimensions. Data governance is not an appendix to the model. It affects the reliability of mobility-derived demand indicators, the credibility of financial forecasts, and the risk profile attached to the final Investment Attractiveness Score.

4.7 Euro-Mediterranean infrastructure integration and regulatory divergence

The Euro-Mediterranean context gives the review a comparative boundary. Infrastructure connectivity remains a policy priority across the Union for the Mediterranean, while regulatory fragmentation and institutional variation continue to shape investment conditions. This matters because Casablanca and EU cities may face comparable urban infrastructure choices but operate under different procurement, data protection, PPP, and enforcement environments.

The literature on Euro-Mediterranean integration often discusses infrastructure at the macro level: connectivity, investment flows, trade integration, competitiveness, and regional development. What remains underdeveloped is a project-level evaluation framework that can compare specific urban infrastructure opportunities across governance environments. This proposal narrows that broader issue to mobility-dependent smart urban infrastructure, using Casablanca as the empirical anchor and EU/Italian regulatory conditions as a comparative benchmark.

4.8 Synthesis of the research gap

The literature converges on a single weakness: the relevant analytical components exist, but they do not yet form a coherent investment-evaluation logic. Smart city research explains technological and governance transformation but often under-specifies financial viability. Mobility analytics estimates flows and exposure but rarely converts them into project-level appraisal. Infrastructure finance offers NPV, ROI, payback and sensitivity tools, yet often treats demand and governance conditions as exogenous assumptions. MCDA offers ranking logic, but frequently uses generic risk scores rather than variables connected to cash-flow uncertainty and regulatory friction. PPP and data governance research clarify contractual and compliance constraints, but rarely translate them into valuation variables.

This synthesis narrows the gap to a precise research problem: existing research does not yet explain how mobility-derived demand, project-level financial viability, regulatory constraints, governance quality and data reliability can be operationalised together in one risk-adjusted evaluation framework for mobility-dependent smart urban infrastructure.

5. Research Gap

5.1 Formal statement of the gap

Despite substantial progress in smart city governance, mobility analytics, infrastructure finance, MCDA, PPP governance and data regulation, no sufficiently integrated framework has yet been established for evaluating mobility-dependent smart urban infrastructure as a project-level investment asset. Previous research has generated valuable but partial tools: smart city studies clarify the technological and institutional role of digital infrastructure; mobility studies estimate traffic, exposure and spatial demand; infrastructure finance measures NPV, ROI, payback and risk-adjusted returns; MCDA ranks alternatives under multiple criteria; PPP research analyses value for money and risk allocation; and data-governance studies examine privacy, access and compliance. However, these streams have not yet been operationalised together into a replicable model that connects spatial demand, cash-flow viability, governance conditions and data reliability in one investment decision framework.

The gap is therefore not that nobody has discussed smart cities, PPPs, mobility data or infrastructure finance. That would be a false gap. The real gap is integrative and operational: current knowledge does not sufficiently explain how mobility-derived demand indicators can be translated into financial appraisal and then adjusted for regulatory risk, governance quality, demand uncertainty and data quality. This matters because mobility-dependent assets - including digital out-of-home panels, smart shelters, connected street furniture and mobility-linked urban nodes - can be misclassified if traffic exposure, financial feasibility and governance risk are evaluated separately.

5.2 Typology of the research gap

Table 2. Typology of the Research Gap and Proposed Response

Gap type	What is missing	Why it matters	How this study addresses it
Knowledge gap	The literature does not sufficiently explain how mobility-derived demand, financial viability, governance risk and data quality interact in one investment decision.	Without this integration, research remains divided between technical, financial and regulatory silos.	The study defines a unified evaluation logic linking Mobility Potential, Financial Viability, Risk Adjustment, Governance Quality and Data Quality.
Methodological gap	Existing methods are often applied separately: GIS/SUMO for mobility, DCF for finance, MCDA for ranking, and legal review for governance.	Separate methods can produce contradictory conclusions about the same infrastructure asset.	The study combines spatial demand estimation, financial modelling, AHP/TOPSIS/COPRAS-style scoring and sensitivity analysis.
Empirical/data gap	There is limited project-level evidence on how mobility exposure can be converted into revenue-sensitive investment assumptions for smart urban assets.	High traffic does not automatically imply investability; exposure must be connected to cash-flow assumptions and uncertainty ranges.	The Casablanca anchor allows the framework to be calibrated around mobility-dependent infrastructure, with conservative assumptions where revenue data is unavailable.
Contextual gap	Euro-Mediterranean investment discussions often remain macro-level and rarely compare project-level urban assets across different regulatory regimes.	Cities may face similar infrastructure choices but operate under different data, procurement and PPP conditions.	The study uses Casablanca as the empirical anchor and EU/Italian regulatory conditions as a comparative benchmark.

Note. The table links each gap type to its methodological consequence and to the specific response designed in this proposal. Source: Author's own elaboration.

5.3 Why the gap matters

This gap is important because the separation of these fields can lead to systematically weak investment decisions. A high-traffic corridor may appear attractive in a mobility analysis but become financially fragile once CAPEX, OPEX, revenue uncertainty, permitting delays, data restrictions and PPP governance risks are included. Conversely, a site with moderate traffic may be more investable if costs are lower, data reliability is stronger and the governance environment is more predictable. The risk is not only analytical; it is practical. Fragmented evaluation can encourage municipalities to prioritise visible or technologically fashionable projects that lack robust investment logic.

Recent institutional evidence reinforces the urgency of this gap. The European Court of Auditors reported that EU smart city initiatives produced tangible solutions but faced fragmentation that limited wider adoption. The OECD identifies smart city data governance as a core condition for turning real-time urban data into effective policy and service delivery. The World Bank PPP guidance treats value for money, affordability, market assessment and risk allocation as central to PPP appraisal. These sources do not solve the present research problem; rather, they confirm why a model that jointly evaluates mobility demand, financial viability, data reliability and governance risk is necessary.

5.4 Direct link to the proposed research

This study fills the identified gap by developing a risk-adjusted Investment Attractiveness Score for mobility-dependent smart urban infrastructure. The proposed score does not claim to predict investment success with certainty. Instead, it provides a transparent and sensitivity-tested structure for comparing selected infrastructure locations or project alternatives according to five linked dimensions: Mobility Potential, Financial Viability, Risk Adjustment, Governance Quality and Data Quality.

The study is deliberately bounded to avoid an overextended claim. It focuses on mobility-dependent smart urban infrastructure rather than all smart city technologies; it uses Casablanca as the empirical anchor rather than claiming full Euro-Mediterranean generalisability; and it uses EU/Italian regulatory conditions as a comparative benchmark rather than attempting to harmonise all legal systems. These boundaries make the gap researchable, measurable and directly aligned with the main research question.

6. Research Objectives

6.1 Main Aim

The objectives are deliberately limited to four operational steps so that the proposal remains measurable and feasible within a Master-level research timeframe. The study does not attempt to evaluate all smart-city technologies or all Euro-Mediterranean infrastructure systems. It focuses on mobility-dependent smart urban infrastructure, with Casablanca as the empirical anchor and EU/Italian regulatory conditions as the comparative benchmark.

Main Aim:

To develop and empirically calibrate a risk-adjusted evaluation framework for mobility-dependent smart urban infrastructure investment by integrating mobility-derived demand indicators, financial viability metrics, regulatory-governance risk variables, and data quality into a composite Investment Attractiveness Score.

6.2 Specific Objectives

To achieve this aim, the study pursues four specific objectives aligned with the research sub-questions and the proposed methodology:

- **Operationalise mobility-derived demand indicators** by converting traffic intensity, pedestrian flow, spatial exposure, accessibility, dwell time, and public transport proximity into a normalised Mobility Potential Index for selected infrastructure locations.
- **Evaluate financial viability** by calculating CAPEX, OPEX, projected revenue, NPV, ROI, and payback period for selected smart urban infrastructure assets under clearly stated demand, cost, and discount-rate assumptions.
- **Quantify risk and governance adjustment variables** by translating demand uncertainty, data quality, regulatory constraints, and PPP governance conditions into measurable risk coefficients within the investment evaluation framework.
- **Calculate and sensitivity-test the Investment Attractiveness Score** by combining mobility, financial, risk, governance, and data-quality indicators to rank selected project alternatives and assess the robustness of the ranking under alternative scenarios.

6.3 Alignment between Questions, Objectives, Methods, and Outputs

The objectives follow a strict mirror logic: each objective answers one research sub-question, uses a defined method, and produces a verifiable output.

Table 3. Alignment between Research Questions, Objectives, Methods, and Outputs

Research sub-question	Specific objective	Method	Expected output
SQ1: How can mobility indicators be converted into demand indicators?	Operationalise mobility-derived demand indicators.	GIS/SUMO, traffic flow, exposure scoring, spatial accessibility assessment.	Mobility Potential Index.
SQ2: Which financial indicators capture viability?	Evaluate financial viability.	CAPEX/OPEX modelling, revenue assumptions, NPV, ROI, payback calculation.	Financial Viability Score.
SQ3: How can risk and governance be measured?	Quantify risk and governance variables.	Regulatory comparison, PPP governance scoring, data-quality assessment.	Risk and Governance Adjustment Variables.
SQ4: Can the IAS rank alternatives robustly?	Calculate and sensitivity-test the IAS.	MCDA weighting, composite scoring, sensitivity analysis.	Ranked alternatives and robustness test.

Note. GIS = geographic information system; SUMO = Simulation of Urban Mobility; DCF = discounted cash flow; IAS = Investment Attractiveness Score. Source: Author's own elaboration.

6.4 SMART and Feasibility Safeguard

These objectives are specific because each targets a defined model component; measurable because each produces an index, score, coefficient, or ranking; achievable because the project uses available mobility, spatial, financial, and regulatory inputs; realistic because it remains limited to selected infrastructure assets and selected locations; and time-constrained because the empirical calibration is bounded to the Master research period rather than an open-ended citywide implementation programme.

7. Conceptual Framework

This section translates the research question into a testable conceptual architecture. The framework explains how mobility-derived demand, financial viability, governance conditions, risk exposure, and data quality jointly shape the investment attractiveness of mobility-dependent smart urban infrastructure. It is designed for assets whose value depends on spatial exposure and urban flow patterns, including digital out-of-home advertising panels, smart bus shelters, connected street furniture, and mobility-linked urban nodes.

7.1 Core variable structure

The dependent variable is the Investment Attractiveness Score (IAS). The IAS is defined as a composite measure of whether a selected infrastructure location or project alternative is financially viable, spatially justified, institutionally feasible, and robust under uncertainty. It is not a guaranteed return forecast; it is a structured comparative score for ranking alternatives under stated assumptions.

Table 4. Conceptual Roles of the Main Variables in the IAS Framework

Conceptual role	Variable	Function in the model
Dependent variable	Investment Attractiveness Score (IAS)	Final composite outcome to be explained, compared, and ranked.
Primary independent variable	Mobility Potential	Captures spatial demand through traffic intensity, pedestrian flow, dwell time, exposure, accessibility, and public transport proximity.
Mediating variable	Expected Revenue / Financial Viability	Converts mobility exposure into investment performance through projected revenue, CAPEX, OPEX, NPV, ROI, and payback period.
Adjustment variables	Risk Exposure and Governance Quality	Adjust the relationship between financial viability and final attractiveness by accounting for uncertainty, regulatory constraints, PPP quality, procurement transparency, and enforcement reliability.
Validity / confidence variable	Data Quality	Determines how reliable the mobility and financial estimates are, based on data completeness, granularity, recency, consistency, and legal usability.

Note. IAS = Investment Attractiveness Score. The table distinguishes dependent, independent, mediating, moderating, and confidence-related variables. Source: Author's own elaboration.

7.2 Theoretical logic of the framework

The framework postulates that Mobility Potential is the upstream demand driver of smart urban infrastructure investment. A location with stronger traffic intensity, pedestrian flow, exposure duration, and accessibility should generate higher potential demand for mobility-dependent assets. However, the relationship between mobility and investment attractiveness is not direct. Mobility becomes economically meaningful only when it can be translated into expected revenue.

The relationship between Mobility Potential and Investment Attractiveness is therefore mediated by Financial Viability. Financial viability captures whether the expected demand can generate enough revenue to justify capital expenditure, operating expenditure, and the investment horizon. This is why the framework uses CAPEX, OPEX, NPV, ROI, and payback period as the financial bridge between spatial exposure and final investment attractiveness.

The framework then introduces Risk Exposure as a negative adjustment mechanism. Risk Exposure includes demand uncertainty, revenue volatility, regulatory friction, data availability risk, and sensitivity to changes in cost or discount-rate assumptions. A project may score highly on mobility and financial viability, but its attractiveness decreases if its revenues are unstable, its data assumptions are weak, or its regulatory exposure is excessive.

Governance Quality functions as an institutional moderator. In public-private partnership contexts, investment attractiveness depends not only on projected cash flows, but also on procurement transparency, contractual clarity, risk allocation, permitting predictability, and enforcement quality. This logic is grounded in PPP theory: efficient risk allocation improves value for money because it creates incentives for parties to manage risks effectively (World Bank, 2024c).

Data Quality operates as a confidence condition rather than a direct revenue driver. If mobility data is incomplete, outdated, inconsistent, or legally restricted, then the estimated relationship between mobility exposure and financial viability becomes less reliable. This is consistent with the OECD view that smart city data governance depends on data availability, effective use, privacy, compliance, security, and business-model conditions (OECD, 2023).

Figure 1 presents the conceptual logic of the study. It shows that mobility potential does not directly determine investment attractiveness; it first contributes to financial viability, while risk exposure, governance quality, and data quality adjust the credibility and robustness of the final Investment Attractiveness Score.

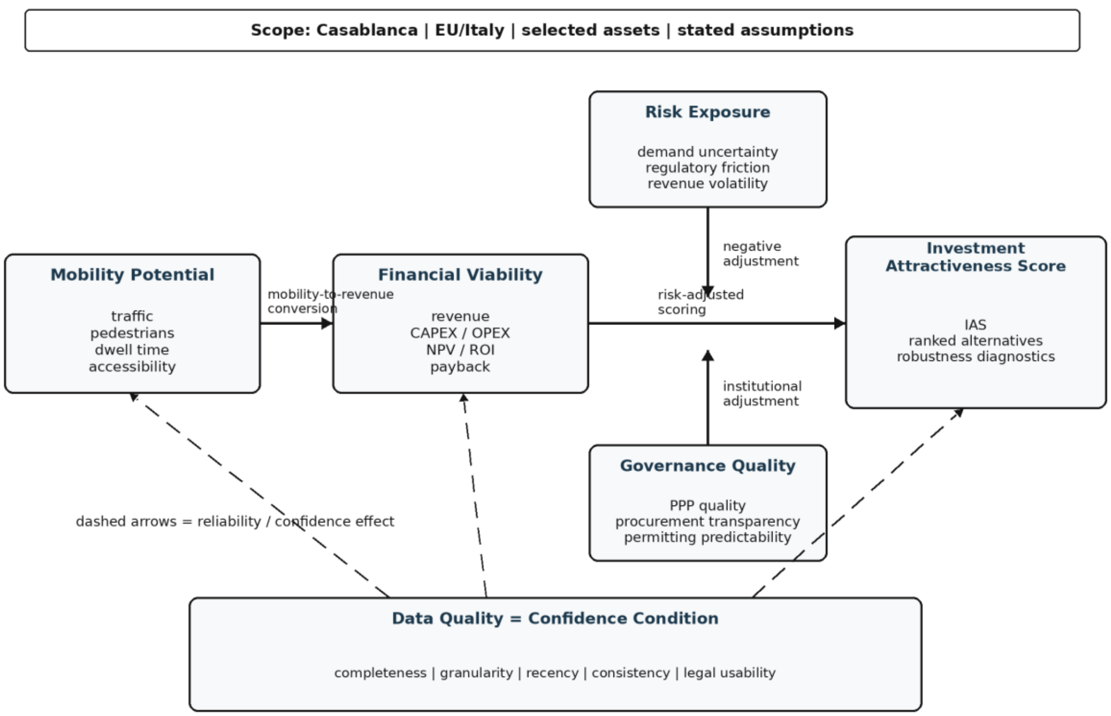

Figure 1. Conceptual Framework for Risk-Adjusted Smart Urban Infrastructure Investment Evaluation.

Note. IAS = Investment Attractiveness Score; MP = Mobility Potential; FV = Financial Viability; RE = Risk Exposure; GQ = Governance Quality; DQ = Data Quality. Solid arrows show the main pathway; dashed arrows show adjustment or confidence effects. Source: Author's own elaboration.

7.3 Decision-theory foundation

The framework uses Multi-Criteria Decision Analysis because the investment decision cannot be reduced to one variable. The Analytic Hierarchy Process supports the weighting of criteria because it was developed as a structured method for deriving ratio-scale priorities from pairwise comparisons (Saaty, 1980). TOPSIS supports the ranking stage because it evaluates alternatives according to their distance from a positive ideal solution and a negative ideal solution (Hwang & Yoon, 1981).

In this study, AHP defines the relative weight of the criteria, while TOPSIS or COPRAS ranks selected infrastructure alternatives. The conceptual variables therefore define what must be measured, and MCDA provides the decision mechanism for weighting, comparing, and ranking those variables.

7.4 Conceptual propositions

Table 5. Conceptual Propositions Underlying the IAS Framework

Proposition	Conceptual relationship
P1 - Mobility-to-demand	Higher Mobility Potential increases expected exposure for mobility-dependent smart urban infrastructure.
P2 - Mediation	The effect of Mobility Potential on Investment Attractiveness is mediated by Financial Viability because exposure must be converted into projected revenue, NPV, ROI, and payback performance.
P3 - Risk adjustment	Higher Risk Exposure weakens Investment Attractiveness by reducing confidence in projected returns.
P4 - Governance moderation	Higher Governance Quality strengthens Investment Attractiveness by improving procurement transparency, PPP risk allocation, permitting predictability, and enforcement credibility.
P5 - Data-quality condition	Higher Data Quality increases confidence in the estimated Mobility Potential and Financial Viability scores, while weak data quality reduces the reliability of the final IAS.

Note. The propositions are conceptual rather than causal claims; they structure the logic to be tested through scoring and sensitivity analysis. Source: Author's own elaboration.

7.5 Delimitation of the conceptual framework

The framework is deliberately bounded. It does not attempt to explain the full performance of all smart city technologies. It focuses only on mobility-dependent smart urban infrastructure assets whose value depends on spatial exposure, investment cost, governance conditions, and data reliability.

The framework excludes variables that would require a separate research design, including citizen satisfaction, long-term environmental impact, behavioural acceptance, full macroeconomic spillovers, and political ideology. These variables are relevant to smart city evaluation, but including them would dilute the model and make the project empirically unmanageable. The focus remains on investment attractiveness, not total social welfare.

8. Methodology

8.1 Research Design

This research adopts a sequential mixed-methods design combining spatial mobility analysis, project-level financial modelling, regulatory-governance assessment, and multi-criteria decision analysis. The design is sequential because the final Investment Attractiveness Score cannot be observed directly; it must be built step by step from auditable intermediate outputs. Mobility analysis estimates exposure. Financial modelling converts exposure assumptions into investment viability. Regulatory-governance assessment captures implementation constraints. MCDA integrates these heterogeneous indicators into a ranked decision output.

This design is preferable to a purely qualitative legal analysis because the research question requires comparable site-level scores. It is also preferable to a purely financial appraisal because NPV and ROI alone cannot capture data quality, demand volatility, procurement uncertainty, or PPP governance risk. The methodology therefore follows five linked stages:

- spatial and mobility assessment to estimate traffic exposure and location attractiveness;
- financial modelling to convert exposure assumptions into CAPEX, OPEX, revenue, NPV, ROI, and payback indicators;
- regulatory and governance scoring to assess PPP feasibility, data-governance constraints, and implementation risk;
- multi-criteria scoring to calculate the Investment Attractiveness Score;
- sensitivity and robustness testing to determine whether rankings remain stable under alternative assumptions.

This architecture reduces the black-box risk: each stage produces a measurable output that can be inspected before it enters the final score.

8.2 Research Scope and Case Selection

The empirical scope is deliberately bounded. The study focuses on mobility-dependent smart urban infrastructure assets in Casablanca, including digital out-of-home advertising panels, smart bus shelters, connected street furniture, and mobility-linked urban nodes. These assets are selected because their investment value depends directly on spatial exposure, traffic flow, operating cost, and regulatory feasibility.

Casablanca is selected as the empirical anchor for three reasons. First, it is a high-pressure urban mobility environment where infrastructure placement decisions depend strongly on exposure and accessibility. Second, the researcher has prior applied experience with Casablanca-based traffic modelling, GIS analysis, and advertising-infrastructure profitability assessment. Third, Casablanca provides a relevant emerging-market context for testing how mobility-derived demand can be integrated with financial and governance indicators.

The European Union and Italy are used as regulatory benchmarks rather than full parallel empirical cases. This choice keeps the project feasible: the study compares regulatory logic, procurement structure, PPP governance, and data-protection constraints without attempting to run a full multi-city econometric comparison.

8.3 Population and Sampling

The unit of analysis is not an individual respondent. The primary unit of analysis is the infrastructure location or project alternative. The proposed empirical sample consists of 5 to 10 candidate sites in Casablanca, selected through purposive sampling.

Candidate sites will be selected according to four inclusion criteria:

- high or medium traffic intensity;
- proximity to public transport, major road corridors, or high-flow commercial zones;

- practical relevance for advertising or smart street-furniture deployment;
- availability of spatial, mobility, cost-estimation, and regulatory information.

A sample of 5 to 10 sites is methodologically defensible because the aim is to develop and calibrate a proof-of-concept evaluation framework, not to produce a complete city-wide investment map. This scope allows detailed site-level modelling while remaining feasible within a Master's-level research timeline.

If AHP expert weighting is feasible, the target expert sample will include 6 to 10 purposively selected participants from urban planning, infrastructure finance, PPP/procurement, smart-city operations, advertising infrastructure, or applied urban economics. If expert access is limited, the study will use a documented baseline weighting scheme derived from the literature and will test alternative weight scenarios through sensitivity analysis.

8.4 Data Collection Instruments

8.4.1 Spatial and Mobility Data

Spatial and mobility data will be collected from available GIS layers, road-network data, public transport proximity data, municipal or open-source traffic datasets, and SUMO simulation outputs where observed traffic counts are unavailable. QGIS will be used to process coordinates, road-network proximity, transport accessibility, and spatial exposure indicators. This choice is justified because QGIS is an official open-source GIS environment designed for spatial data handling, mapping, and analysis. SUMO will be used for simulated mobility flows where required because it is an open-source microscopic and continuous multi-modal traffic simulation package designed to handle large networks.

Table 6. Mobility Indicators Used to Construct the Mobility Potential Index

Indicator	Measurement logic	Output
Traffic intensity	Observed or simulated vehicle flow / peak-hour traffic	Traffic score
Pedestrian exposure	Observed count, proxy estimate, or site-level pedestrian intensity	Pedestrian score
Dwell time	Estimated slowing, stopping, or waiting duration near the site	Dwell-time score
Public transport proximity	Distance to tram, bus, interchange, or high-flow mobility node	Accessibility score
Spatial visibility	Line-of-sight, road corridor exposure, or frontage proxy	Visibility score
Location accessibility	Network centrality and commercial-area proximity	Accessibility score

Note. Indicators may be observed, simulated, or proxied depending on data availability. Source: Author's own elaboration.

These indicators will be normalised to construct the Mobility Potential Index.

8.4.2 Financial Data

Financial data will include estimated CAPEX, OPEX, projected revenue, discount-rate assumptions, payback period, NPV, and ROI. CAPEX covers equipment, installation, permitting, infrastructure preparation, and deployment costs. OPEX covers maintenance, energy, connectivity, supervision, repairs, and data-management costs. Where supplier quotations or operator-level revenue data are unavailable, the study will use conservative assumptions recorded in an assumption register.

The core revenue logic is:

$$\textit{Annual Revenue} = \textit{Daily Traffic Exposure} \times \textit{Exposure Coefficient} \times \textit{Revenue Rate} \times \textit{Operating Days}$$

The exposure coefficient adjusts raw traffic volume for viewability, dwell time, visibility angle, and site relevance. The revenue rate represents the assumed monetisation rate, such as advertising revenue per thousand impressions or an equivalent concession/rental value. This formula prevents the model from assuming that traffic automatically becomes revenue.

8.4.3 Regulatory and Governance Data

Regulatory and governance data will be collected through documentary analysis of procurement rules, PPP frameworks, municipal advertising regulations, data-protection requirements, and institutional guidance. This component is necessary because PPP evaluation depends on value-for-money logic, risk allocation, affordability, market assessment, procurement transparency, and enforceability. The World Bank's PPP guidance treats risk allocation and value for money as central to PPP appraisal.

Table 7. Governance and Regulatory Scoring Rubric

Dimension	Scoring criterion	Scale
PPP framework clarity	Roles, contractual responsibilities, and risk allocation are clearly defined	0-5
Procurement transparency	Tendering and award criteria are public, comparable, and auditable	0-5
Permitting predictability	Approval steps are defined, time-bounded, and administratively predictable	0-5
Data-protection constraint	Data rules affect usable granularity, consent, storage, or monetisation	0-5
Enforcement reliability	Contractual and regulatory obligations can be enforced credibly	0-5
Compliance burden	Compliance increases cost, delay, or operating complexity	0-5

Note. Each dimension is scored from 0 to 5, where 0 indicates absence or severe weakness and 5 indicates strong clarity, reliability, or usability. Source: Author's own elaboration.

The scores are not presented as legal truth. They operationalise regulatory and governance conditions as investment-relevant risk variables. Each score must be justified by a cited document or an explicitly stated assumption.

8.4.4 Data Quality Assessment

Data quality will be evaluated because mobility-based investment scoring is only as reliable as the data used to construct it. The data quality score will assess completeness, spatial granularity, temporal recency, consistency across sources, and legal usability. This is consistent with OECD work on smart city data governance, which identifies data management, sharing, compliance, privacy, security, and business-model challenges as central constraints for data-enabled urban systems.

8.5 Data Analysis Procedure

Stage 1 - Spatial data processing.

All selected candidate sites will be geocoded in QGIS. The analysis will calculate proximity to major roads, public transport nodes, high-density zones, and commercial corridors. Where observed traffic counts are unavailable, SUMO simulation outputs will be used to approximate traffic flow. The output is a standardised site-level spatial and mobility dataset.

Stage 2 - Mobility Potential Index.

Mobility indicators will be normalised to a 0-1 scale. For benefit criteria, higher values indicate stronger investment potential. For constraint criteria, values will be reversed where necessary. The Mobility Potential Index is calculated as:

$$MPI = w1(Traffic) + w2(Pedestrian\ Flow) + w3(Dwell\ Time) + w4(Visibility) + w5(Accessibility)$$

Stage 3 - Financial viability modelling.

For each site, the study will estimate CAPEX, annual OPEX, annual revenue, net cash flow, payback period, ROI, and NPV. The NPV formula is:

$$NPV = \Sigma\ [CFt / (1 + r)^t] - I0$$

where CFt is annual net cash flow, r is the discount rate, and I0 is initial investment. The analysis will use a 5- to 10-year investment horizon depending on the assumed concession period.

Stage 4 - Governance and risk scoring.

Regulatory and governance documents will be coded into measurable scores. This stage produces a Governance Quality Score, Regulatory Risk Score, Data-Protection Constraint Score, Compliance Burden Score, and Data Quality Score. To avoid double counting, governance quality measures institutional support and clarity, while risk exposure measures downside uncertainty, volatility, and constraints that can reduce projected returns.

Stage 5 - MCDA weighting and ranking.

The study will use AHP to structure criteria weights and TOPSIS or COPRAS to rank selected alternatives. AHP is used because it provides a structured method for deriving ratio-scale priorities through pairwise comparisons. TOPSIS is used because it ranks alternatives according to their closeness to a positive ideal solution and distance from a negative ideal solution. The Investment Attractiveness Score follows the structure:

$$IAS = w1(MP) + w2(FV) - w3(RE) + w4(GQ) + w5(DQ)$$

where MP is Mobility Potential, FV is Financial Viability, RE is Risk Exposure, GQ is Governance Quality, and DQ is Data Quality.

Stage 6 - Sensitivity and robustness testing.

Sensitivity analysis will test whether rankings change under alternative assumptions. If a site ranks highly only under optimistic assumptions, the model will report it as fragile rather than genuinely robust.

Table 8. Parameters for Sensitivity Analysis

Parameter	Sensitivity range

Parameter	Sensitivity range
Traffic exposure	+/- 20%
Revenue conversion rate	+/- 25%
CAPEX	+/- 15%
OPEX	+/- 15%
Discount rate	+/- 2 to 3 percentage points
Governance score	low / medium / high
Data quality score	low / medium / high

Note. Ranges are preliminary and may be refined after data availability and empirical calibration are assessed. Source: Author's own elaboration.

8.6 Software and Analytical Tools

Table 9. Software and Analytical Tools Used in the Study

Task	Tool
Spatial data cleaning and mapping	QGIS
Traffic simulation / mobility modelling	SUMO
Data cleaning and calculation	Python/pandas or R
Financial modelling	Excel or Python
MCDA scoring	Excel, Python, or R
Sensitivity analysis	Excel, Python, or R
Assumption documentation	Assumption register table

Note. Tool choice may be adjusted if equivalent reproducible software is required by the university or supervisor. Source: Author's own elaboration.

8.7 Methodological Workflow

Figure 2 translates the research design into a reproducible methodological sequence. It shows how selected Casablanca sites move from spatial and mobility data collection to financial modelling, governance scoring, MCDA integration, and sensitivity testing before producing ranked investment alternatives.

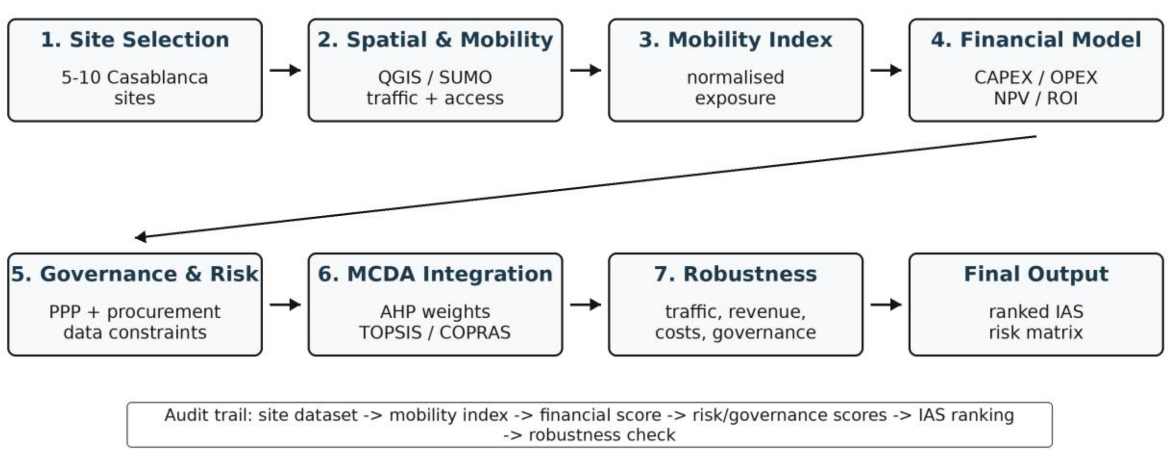

Figure 2. Methodological Workflow for Developing and Testing the Investment Attractiveness Score.

Note. MCDA = multi-criteria decision analysis; GIS = geographic information system; SUMO = Simulation of Urban Mobility; DCF = discounted cash flow; IAS = Investment Attractiveness Score. The workflow is sequential and ends with ranked alternatives, a profitability-risk matrix, and robustness diagnostics. Source: Author's own elaboration.

8.8 Validity, Reliability, and Reproducibility

The methodology includes four safeguards. First, all indicators will be defined before scoring. Second, all assumptions will be recorded in an assumption register. Third, all variables will be normalised using a stated procedure. Fourth, sensitivity analysis will test whether the IAS is stable under alternative plausible assumptions.

Reproducibility will be supported through a data-source inventory, a variable dictionary, a normalisation protocol, a scoring rubric, a financial-assumption register, a sensitivity-analysis table, and an MCDA calculation sheet. These materials ensure that another researcher could reproduce the same evaluation framework if provided with the same data and assumptions.

8.9 Ethical and Data Governance Considerations

The study will use aggregated and anonymised mobility data only. It will not collect individual-level GPS traces or personally identifiable movement data. Where simulated mobility data is used, the model will clearly distinguish between observed data and simulation outputs. The IAS will be presented as a decision-support tool rather than a neutral substitute for public deliberation, legal due diligence, or equity analysis.

9. Data Requirements and Data Management

9.1 Data Strategy and Unit of Analysis

This research uses a structured combination of secondary spatial data, simulated mobility data, documentary regulatory data, financial assumption data, and, where feasible, expert-based weighting data. The study does not rely on individual-level personal mobility records. The primary unit of analysis is the infrastructure location or project alternative, not individual citizens, households, or identifiable road users.

The empirical component will focus on 5 to 10 candidate sites in Casablanca. These sites will be selected to test the proposed Investment Attractiveness Score across different exposure conditions, including high-traffic corridors, public transport nodes, commercial zones, and mixed-use urban locations. EU/Italian regulatory conditions will be used as a benchmark for governance and data-protection comparison, not as a second full empirical case study.

The data strategy follows one reproducibility rule: every variable included in the Investment Attractiveness Score must have a defined source, measurement logic, quality-control procedure, and fallback plan if direct data are unavailable.

9.2 Data Categories and Required Variables

Table 10. Data Categories, Variables, Sources, and Model Roles

Data category	Variables required	Preferred source	Role in the model
Spatial and road-network data	Site coordinates; road	OpenStreetMap; QGIS layers;	Defines the physical location,

Data category	Variables required	Preferred source	Role in the model
	hierarchy; intersections; road density; public transport proximity; land-use context	municipal maps; satellite/urban atlas data	accessibility, and exposure context of each candidate site
Mobility data	Traffic intensity; peak/off-peak flow; pedestrian-flow proxy; dwell-time proxy; exposure corridor; directional flow	Municipal traffic data where available; SUMO simulation where unavailable; limited field observation if feasible	Constructs the Mobility Potential Index
Urban and demographic data	Population density; urban density; commercial density; land-use mix; public transport integration	HCP Morocco; World Bank WDI; OpenStreetMap POIs; available urban datasets	Refines location attractiveness and demand context
Financial data	CAPEX; OPEX; maintenance; electricity/connectivity; installation; lease/concession fees; revenue assumptions	Supplier documentation; public tenders; concession documents; industry benchmarks; assumption register	Constructs the Financial Viability Score
Macroeconomic data	Inflation; exchange-rate assumptions; discount-rate proxy; country-risk context	World Bank WDI; IMF; Eurostat; OECD Data Explorer	Supports NPV, ROI, discount-rate, and scenario assumptions
Regulatory and governance data	PPP framework; procurement transparency; permit process; data-protection rules; advertising restrictions; enforcement quality	Moroccan laws; CNDP/Loi 09-08; EU/GDPR; Italian procurement rules; World Bank PPP resources; OECD governance materials	Constructs Governance Quality and Risk Adjustment variables
Data quality metadata	Completeness; granularity; recency; consistency; legal usability; source reliability	Researcher-generated data-quality register	Determines confidence attached to each site-level score

Note. OSM = OpenStreetMap; SUMO = Simulation of Urban Mobility; CAPEX = capital expenditure; OPEX = operating expenditure; NPV = net present value; ROI = return on investment. Source: Author's own elaboration.

OpenStreetMap is appropriate for base geographic and road-network extraction because it creates and distributes free geographic data. QGIS is used for spatial processing because it is the official open-source GIS environment used for mapping, spatial analysis, and geodata handling. SUMO is used only where simulation is required because it is an open-source, microscopic and continuous multi-modal traffic simulation package designed for large networks (Eclipse Foundation, 2026; OpenStreetMap Wiki, n.d.; QGIS Project, n.d.).

9.3 Source Hierarchy and Reliability Rules

The study applies a strict source hierarchy to reduce bias and avoid treating weak assumptions as evidence. Direct official data will always be preferred over simulation or benchmark assumptions.

Table 11. Source Reliability Hierarchy for Data Collection

Reliability rank	Source type	Use rule
1 - Highest reliability	Official municipal traffic counts; procurement documents; legal texts; national statistical offices	Used directly where available
2 - High reliability	International institutional databases: World Bank, IMF, Eurostat, OECD	Used for macroeconomic, demographic, governance, and benchmark indicators
3 - Medium reliability	OpenStreetMap; GIS layers; satellite/urban atlas data	Used for spatial proxies and network structure
4 - Conditional reliability	SUMO simulation outputs	Used only when observed traffic data are

Reliability rank	Source type	Use rule
		unavailable or incomplete
5 - Assumption-based	Market benchmarks; supplier quotations; estimated revenue rates	Used only with explicit low/base/high sensitivity ranges

Note. Lower-ranked sources are not excluded; they are used with stronger documentation and sensitivity testing. Source: Author's own elaboration.

The World Bank describes WDI as its primary collection of development indicators compiled from officially recognised international sources. OECD Data Explorer provides access to OECD statistical data. These sources are therefore appropriate for macroeconomic, demographic, and comparative indicators.

9.4 Sampling Strategy

The study uses purposive stratified site sampling. The target sample is 5 to 10 candidate infrastructure locations in Casablanca. The sample is not intended to be statistically representative of all Casablanca infrastructure locations. It is designed as a proof-of-concept analytical sample to test whether the Investment Attractiveness Score can rank alternatives transparently and robustly.

Inclusion criteria:

- the site can be geolocated accurately using coordinates;
- the site is located near a road corridor, public transport node, commercial corridor, or high-exposure urban space;
- the site is relevant for mobility-dependent infrastructure such as digital advertising panels, smart bus shelters, connected street furniture, or mobility-linked urban nodes;
- at least three core data categories are available: spatial data, mobility proxy, and cost/revenue assumption;
- the site can be compared with other sites using the same scoring framework.

Exclusion criteria:

- the location cannot be geocoded reliably;
- spatial or mobility data are too incomplete for scoring;
- the site involves sensitive locations where visibility analysis could create ethical or privacy concerns;
- the infrastructure use case is not mobility-dependent;
- the site requires confidential data that cannot be replaced by defensible assumptions.

9.5 Expert Data for Weighting

If AHP weighting is used, the study will collect expert judgements from 6 to 10 purposively selected experts. The panel may include urban planners, infrastructure-finance professionals, PPP or procurement specialists, smart-city practitioners, advertising-infrastructure operators, or academic experts in urban economics, transport, and data-driven decision-making.

Experts will be included if they have professional or academic exposure to urban infrastructure, mobility planning, infrastructure finance, PPP governance, or data-driven urban policy. Experts will be excluded if they cannot provide informed judgement on at least one model dimension or if a conflict of interest creates excessive bias. If expert access is not feasible, the study will use a documented baseline weighting scheme derived from the literature and will test alternative weighting scenarios through sensitivity analysis.

9.6 Data Collection Protocol

The collection process follows six reproducible steps:

Step 1 - Candidate site identification. Candidate sites will be selected from Casablanca major corridors, public transport nodes, commercial areas, and mixed-use urban zones. Each site will receive a unique site identifier.

Step 2 - Spatial extraction. Coordinates, road-network structure, public transport proximity, land-use context, and nearby points of interest will be extracted using QGIS and OpenStreetMap. Spatial layers will be projected using a consistent coordinate reference system to avoid measurement distortion.

Step 3 - Mobility estimation. Observed traffic counts will be used where available. Where observed data are unavailable, SUMO simulation outputs will estimate traffic intensity and flow structure. Simulated values will be clearly labelled as simulated rather than observed.

Step 4 - Financial data compilation. CAPEX, OPEX, maintenance costs, connectivity costs, lease/concession assumptions, and revenue assumptions will be compiled into a financial assumption register.

Step 5 - Regulatory and governance coding. Legal and governance documents will be coded using a structured 0-5 scoring rubric covering procurement transparency, PPP risk allocation, permitting predictability, data-protection constraints, enforcement reliability, and compliance burden.

Step 6 - Data quality scoring. Each dataset will receive a Data Quality Score based on completeness, granularity, recency, consistency, and legal usability.

Revenue will not be assumed directly from traffic. It will be estimated through the following logic:

$$\textit{Annual Revenue} = \textit{Daily Exposure} \times \textit{Exposure Coefficient} \times \textit{Revenue Rate} \times \textit{Operating Days}$$

This expression separates physical exposure from monetisable demand and prevents the model from confusing visibility with actual investment value.

9.7 Data Quality Management

Table 12. Data Quality Scoring Criteria

Criterion	Meaning	Score logic
Completeness	Are the required variables available for the site?	0 = missing; 5 = complete
Granularity	Is the data precise enough at site level?	0 = city-level only; 5 = site-level
Recency	Is the data recent enough for investment evaluation?	0 = outdated; 5 = current/recent
Consistency	Do sources agree or conflict?	0 = contradictory; 5 = consistent
Legal usability	Can the data be used ethically and legally?	0 = unusable; 5 = fully usable
Source reliability	Is the source official, institutional, open-source, simulated, or assumption-based?	0 = unverified; 5 = official/primary

Note. Each criterion is scored from 0 to 5; higher values indicate stronger quality, precision, or legal usability. Source: Author's own elaboration.

The final Data Quality Score will be included in the IAS to prevent weak data from producing falsely precise investment rankings.

9.8 Bias Control Procedures

Table 13. Bias Risks and Control Procedures

Bias risk	Control procedure
Traffic measurement bias	Compare observed data, OSM-based spatial structure, and SUMO simulation outputs where possible
Revenue assumption bias	Use conservative, baseline, and optimistic revenue scenarios
Governance scoring subjectivity	Use a predefined coding rubric and documentary justification for each score
Open-source spatial-data incompleteness	Cross-check OSM data against QGIS layers, satellite imagery, and available municipal sources
Expert-weighting bias	Use structured AHP forms and test alternative weighting scenarios
Temporal bias	Record the date and period of each dataset and flag outdated sources

Note. The controls do not eliminate uncertainty; they make uncertainty visible and testable. Source: Author's own elaboration.

Sensitivity analysis will test whether rankings remain stable when uncertain variables change. The most fragile variables are expected to be revenue rate, exposure coefficient, traffic volume, governance score, and data quality score.

9.9 Ethical and Storage Protocol

The study will use aggregated and anonymised data only. It will not collect individual GPS traces, personally identifiable mobility records, licence plates, advertising identifiers, or personal contact data from road users. Under European data-protection guidance, irreversibly anonymised data that no longer identifies individuals falls outside the category of personal data. For Morocco, the project will respect the logic of Loi 09-08, which concerns the protection of individuals in relation to personal data processing and is supervised by the CNDP (Commission Nationale de Controle de la Protection des Donnees a Caractere Personnel, n.d.; European Commission, n.d.).

Raw data and derived datasets will be stored in a structured project folder with read-only raw files, cleaned datasets, transformation logs, a variable dictionary, an assumption register, version-controlled analysis scripts, and anonymised expert-response files if expert weighting is used. Files containing expert responses will be anonymised and stored separately from any contact information. Access will be restricted to the researcher and academic supervisor.

9.10 Reproducibility Package

To ensure that another researcher can reproduce the study, the final research package will include:

- a site list with inclusion and exclusion justification;

- a data-source inventory;
- a variable dictionary;
- a data-quality scoring table;
- a financial assumption register;
- QGIS processing notes;
- SUMO scenario assumptions if simulation is used;
- AHP weighting sheet or alternative weighting scenario table;
- MCDA calculation sheet;
- sensitivity-analysis outputs.

Python/pandas or R will be used for data cleaning, calculation, and reproducibility. pandas is an open-source Python library for data analysis and manipulation, while R is a statistical computing and graphics environment (R Core Team, n.d.; The pandas development team, n.d.).

10. Proposed Analytical Model

This section specifies the analytical mechanism used to convert heterogeneous spatial, financial, governance, risk, and data-quality indicators into a transparent Investment Attractiveness Score. The model is deliberately designed as a multi-criteria decision model rather than a regression model. A regression design would require a substantially larger sample of observed infrastructure investments and realised revenues; this study instead evaluates 5 to 10 candidate project alternatives and ranks them under explicit assumptions. The appropriate analytical logic is therefore structured scoring, weighting, ranking, and robustness testing, not causal inference from a large-N statistical sample.

The model has three purposes. First, it translates the conceptual framework into measurable variables. Second, it prevents the analysis from confusing visibility with investability by separating mobility exposure from financial viability. Third, it makes uncertainty explicit by testing how rankings change when traffic, revenue, cost, governance, and data-quality assumptions vary.

10.1 Model Specification

The unit of analysis is candidate project alternative i, defined as one infrastructure location or project configuration in Casablanca. The dependent analytical outcome is the Investment Attractiveness Score, written as IAS(i), scaled from 0 to 100. IAS(i) is not a realised financial return and must not be interpreted as a forecasted profit. It is a composite decision score indicating how attractive a project alternative is relative to other alternatives under stated assumptions.

The baseline model is specified as follows:

$$IAS(i) = 100 \times [wMP \times MP(i) + wFV \times FV(i) + wR \times (1 - RE(i)) + wGQ \times GQ(i) + wDQ \times DQ(i)]$$

Subject to:

$$wMP + wFV + wR + wGQ + wDQ = 1; \text{ and } 0 \leq MP(i), FV(i), RE(i), GQ(i), DQ(i) \leq 1.$$

The term (1 - RE(i)) is used because Risk Exposure is a cost-type criterion: higher risk reduces investment attractiveness. This specification avoids negative scoring artefacts while preserving the theoretical logic that risk penalises the final score.

Table 14. Analytical Variables in the Investment Attractiveness Model

Symbol	Variable	Role in the model	Expected direction

Symbol	Variable	Role in the model	Expected direction
IAS(i)	Investment Attractiveness Score	Dependent composite output and ranking variable	Higher = more attractive
MP(i)	Mobility Potential	Primary demand driver based on traffic, pedestrian flow, dwell time, visibility, accessibility, and public transport proximity	Positive
FV(i)	Financial Viability	Mediator translating exposure into investment performance through revenue, CAPEX, OPEX, NPV, ROI, and payback	Positive
RE(i)	Risk Exposure	Adjustment variable capturing demand uncertainty, revenue volatility, regulatory constraints, compliance burden, and macro sensitivity	Negative
GQ(i)	Governance Quality	Institutional moderator capturing PPP quality, procurement transparency, permitting predictability, and enforcement reliability	Positive
DQ(i)	Data Quality	Confidence variable capturing completeness, granularity, recency, consistency, legal usability, and source reliability	Positive
w	Criterion weight	Relative importance of each dimension derived through AHP or documented baseline weighting	Weights sum to 1

Note. IAS(i) refers to the Investment Attractiveness Score for site i; MP = Mobility Potential; FV = Financial Viability; RE = Risk Exposure; GQ = Governance Quality; DQ = Data Quality. Source: Author's own elaboration.

10.2 Variable Measurement

Each dimension is constructed from observable or coded sub-indicators. Continuous indicators are normalised to a 0-1 scale. Ordinal governance and data-quality indicators are scored through predefined rubrics, then normalised where necessary. The model distinguishes benefit criteria, where higher values improve attractiveness, from cost criteria, where higher values reduce attractiveness.

Table 15. Measurement Logic for Composite Variables

Composite variable	Sub-indicators	Measurement / scoring logic	Data source type
Mobility Potential (MP)	Traffic intensity; pedestrian-flow proxy; dwell time; spatial visibility; accessibility; public transport proximity	Normalised 0-1 score; higher values indicate stronger exposure potential	QGIS; OpenStreetMap; municipal traffic data; SUMO outputs; field/proxy observation where feasible
Financial Viability (FV)	Projected annual revenue; CAPEX; OPEX; NPV; ROI; payback period	Composite 0-1 score; NPV and ROI are benefit criteria; CAPEX, OPEX, and payback are cost criteria	Supplier data; public tenders; concession assumptions; assumption register
Risk Exposure (RE)	Demand uncertainty; revenue volatility; regulatory constraints; compliance burden; macroeconomic	Cost-type score from 0 to 1; higher values indicate stronger downside risk	Sensitivity scenarios; regulatory coding; macro assumptions

Composite variable	Sub-indicators	Measurement / scoring logic	Data source type
	sensitivity		
Governance Quality (GQ)	PPP clarity; procurement transparency; permitting predictability; risk-allocation clarity; enforcement reliability	0-5 documentary rubric converted to 0-1; higher values indicate stronger institutional feasibility	Legal texts; public procurement documents; PPP guidance; institutional sources
Data Quality (DQ)	Completeness; granularity; recency; consistency; legal usability; source reliability	0-5 data-quality rubric converted to 0-1; higher values indicate more reliable evidence	Data-source inventory and quality-control checklist

Note. Benefit criteria increase attractiveness; cost or risk criteria reduce attractiveness after reversal or negative adjustment. Source: Author's own elaboration.

10.3 Sub-Index Construction

The model is built through sub-indices rather than raw variables. This is necessary because the indicators have different units: vehicle counts, financial values, percentages, ordinal governance scores, and data-quality ratings cannot be added directly. Each indicator must first be transformed onto a common scale.

For benefit criteria, the normalisation rule is:

$$xnorm(i,j) = [x(i,j) - \min x(j)] / [\max x(j) - \min x(j)]$$

For cost criteria, the normalisation rule is:

$$xnorm(i,j) = [\max x(j) - x(i,j)] / [\max x(j) - \min x(j)]$$

Where x(i,j) is the observed value of alternative i on criterion j, and xnorm(i,j) is the normalised score. This procedure ensures that all final sub-indices follow the same interpretation: higher scores are better, except for Risk Exposure, which is entered into the final IAS as (1 - RE(i)).

Financial Viability will be calculated as a weighted combination of NPV, ROI, and payback performance:

$$FV(i) = aNPV \times NPVnorm(i) + aROI \times ROInorm(i) + aPB \times Paybacknorm(i)$$

Paybacknorm(i) is reverse-normalised because shorter payback is preferred. CAPEX and OPEX enter the model through the cash-flow calculation and may also be included as cost criteria in the MCDA ranking if the evaluation requires direct cost comparison.

Projected annual revenue is estimated through the following transparent conversion formula:

$$Annual\ Revenue(i) = Daily\ Exposure(i) \times Exposure\ Coefficient(i) \times Revenue\ Rate(i) \times Operating\ Days$$

This formula is central to the model. It explicitly separates physical visibility from monetisable demand. Daily exposure measures potential audience; the exposure coefficient adjusts for viewability, dwell time, visibility angle, and site suitability; the revenue rate reflects conservative, baseline, and optimistic monetisation assumptions; and operating days represent the annual active revenue period.

10.4 Weighting Strategy

Weights will be assigned through the Analytic Hierarchy Process (AHP) if expert participation is feasible. AHP is appropriate because it converts structured pairwise comparisons into ratio-scale weights, allowing expert judgement to be formalised transparently. If expert participation is not feasible, the study will use a documented baseline weighting scenario and test alternative weights through sensitivity analysis. Sensitivity analysis is included because MCDA outputs can be sensitive to weight and performance-value uncertainty (Broekhuizen et al., 2015; Wieckowski & Salabun, 2023).

The AHP procedure will follow five steps:

- define the criteria hierarchy: MP, FV, RE, GQ, and DQ;
- collect pairwise comparisons from 6 to 10 experts or construct a literature-based baseline matrix;
- derive the priority vector for criteria weights;
- check the consistency of pairwise judgements; and
- use the final weights in IAS calculation and MCDA ranking.

AHP weights will not be treated as objective truth. They represent a documented preference structure. Because weights can alter rankings, the final model will test alternative weight scenarios: finance-dominant, mobility-dominant, risk-averse, governance-dominant, and equal-weight baselines.

10.5 Ranking Algorithm: TOPSIS and COPRAS Robustness Logic

The baseline IAS provides a transparent composite score. TOPSIS will then be used as the primary ranking method because it ranks each alternative according to its closeness to a positive ideal solution and its distance from a negative ideal solution. This logic fits the research problem because the ideal smart infrastructure alternative should maximise mobility potential, financial viability, governance quality, and data quality while minimising risk exposure.

The TOPSIS procedure will be implemented as follows:

- construct the decision matrix using the selected sites and criteria;
- normalise each criterion so that all indicators are comparable;
- apply AHP or baseline weights to create the weighted normalised matrix;
- identify the positive ideal solution and negative ideal solution;
- calculate each alternative distance from both ideal points;
- calculate relative closeness to the positive ideal solution; and
- rank alternatives from highest to lowest relative closeness.

COPRAS will be used as a robustness check where appropriate because it explicitly handles benefit and cost criteria and can express relative utility among alternatives. The objective is not to multiply methods for appearance; it is to test whether the ranking is method-dependent. If TOPSIS and COPRAS produce substantially different rankings, the result will be reported as unstable rather than presented as a definitive ordering.

10.6 Control Variables and Confounding Risks

Because this model is not a causal regression, the term control variable is used in a design sense: variables that could distort the interpretation of investment attractiveness must either be included in the scoring system, held constant through scope conditions, or tested through sensitivity analysis.

Table 16. Potential Confounders and Treatment in the Analytical Model

Potential confounder	Why it matters	Treatment in the model
Advertising market depth / revenue rate	Traffic exposure may not convert into monetisable demand if advertiser demand or CPM assumptions are weak	Modelled through low/base/high revenue scenarios and sensitivity analysis
Site installation complexity	A visible site may require expensive installation, permits, utilities, or structural preparation	Captured through CAPEX, permitting score, and compliance burden
Land-use and commercial context	High traffic in a low-commercial corridor may be less valuable than moderate traffic in a retail district	Included through location attractiveness and commercial-density proxies
Seasonality and peak-time variation	Traffic may vary by time, day, or season, affecting revenue reliability	Captured through demand uncertainty and sensitivity testing
Data reliability	Weak data may create false precision in the IAS	Captured through the Data Quality Score and data-source hierarchy
Regulatory comparability	EU/Italian and Casablanca conditions may not be directly equivalent	Handled by treating EU/Italy as benchmark conditions, not as a second empirical sample

Note. Confounders are handled through model variables, control logic, and sensitivity scenarios rather than ignored. Source: Author's own elaboration.

10.7 Model Assumptions and Diagnostics

The analytical model rests on explicit assumptions. These assumptions will be tested or documented before the final IAS is interpreted.

Table 17. Model Assumptions and Diagnostic Safeguards

Assumption	Risk if violated	Diagnostic / safeguard
Comparable alternatives	Sites with fundamentally different use cases cannot be ranked meaningfully	Apply inclusion/exclusion criteria and classify infrastructure type before scoring
Correct criterion direction	A cost variable may be incorrectly treated as a benefit variable	Predefine each criterion as benefit or cost before normalisation
No uncontrolled double counting	Governance and risk may penalise the same issue twice	Separate governance quality from downside risk and document each sub-indicator
Reasonable weight structure	AHP or baseline weights may dominate rankings	Run equal-weight, finance-dominant, mobility-dominant, risk-averse, and governance-dominant scenarios
Revenue assumptions are uncertain	IAS may reflect optimistic monetisation rather than investability	Use conservative/base/optimistic revenue rates and report ranking stability
Data quality affects confidence	Low-quality data may produce precise but misleading scores	Apply Data Quality Score and flag low-confidence alternatives
Criteria are not perfectly redundant	Highly correlated indicators may inflate one dimension	Check criterion overlap qualitatively; merge or remove redundant indicators

Note. The safeguards are designed to prevent overinterpretation of a composite score as a deterministic investment decision. Source: Author's own elaboration.

The model does not rely on assumptions required for ordinary least squares regression, such as normally distributed residuals or homoscedasticity, because it is not estimating regression coefficients. Its validity instead depends on MCDA assumptions: correct criterion definition, transparent normalisation, defensible weights, comparable alternatives, and robustness of rankings under alternative assumptions.

10.8 Robustness Checks

Robustness testing is mandatory because the IAS includes uncertain assumptions. The following checks will be performed:

- one-way sensitivity analysis on traffic exposure, revenue rate, CAPEX, OPEX, discount rate, governance score, and data-quality score;
- scenario analysis using conservative, baseline, and optimistic revenue assumptions;
- weight sensitivity using equal-weight, finance-dominant, mobility-dominant, risk-averse, and governance-dominant scenarios;
- method comparison between direct IAS ranking, TOPSIS ranking, and COPRAS ranking where feasible;
- rank-stability assessment using changes in position across scenarios; and
- low-data-quality flagging to prevent weak evidence from being interpreted as robust investment attractiveness.

A ranking will be considered robust only if the same site remains in the top tier across plausible changes in key assumptions. If the top-ranked site changes under small assumption shifts, the result will be reported as fragile. The model therefore prioritises transparent uncertainty over false precision.

10.9 Alignment with Research Questions

Table 18. Alignment between Research Sub-questions and Analytical Outputs

Research sub-question	Analytical mechanism	Model output
SQ1: How can mobility indicators be converted into demand indicators?	MP sub-index and exposure coefficient	Mobility Potential Index
SQ2: Which financial indicators capture viability?	DCF model using revenue, CAPEX, OPEX, NPV, ROI, and payback	Financial Viability Score
SQ3: How can risk and governance be measured?	Risk rubric, governance rubric, data-quality register	Risk Exposure, Governance Quality, and Data Quality scores
SQ4: Can the IAS rank alternatives transparently and robustly?	Weighted IAS, TOPSIS/COPRAS ranking, sensitivity analysis	Ranked alternatives and robustness classification

Note. This table prevents the analytical model from drifting away from the stated research questions. Source: Author's own elaboration.

10.10 Reproducibility Outputs

The final analytical package will include a variable dictionary, raw-to-cleaned data transformation log, normalisation sheet, financial assumption register, AHP or baseline weighting matrix, TOPSIS/COPRAS calculation sheet, sensitivity-analysis table, and final site-ranking matrix. These artefacts are required because the model must be auditable: a reader must be able to trace each final score back to its source data, transformation rule, weight, and assumption.

Figure 3 exposes the internal mechanics of the analytical model. It clarifies how the IAS is built from normalised sub-indices, weighted through MCDA logic, ranked through TOPSIS or COPRAS, and stress-tested through sensitivity analysis.

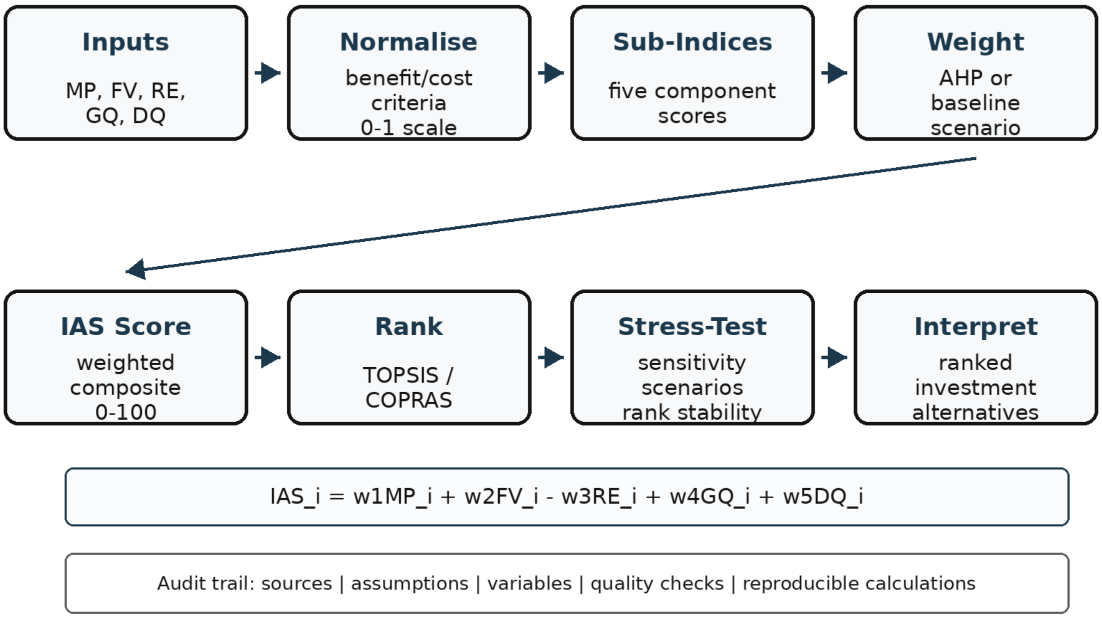

Figure 3. Analytical Workflow for Calculating, Ranking, and Stress-Testing the Investment Attractiveness Score.

Note. IAS = Investment Attractiveness Score; MP = Mobility Potential; FV = Financial Viability; RE = Risk Exposure; GQ = Governance Quality; DQ = Data Quality; AHP = Analytic Hierarchy Process; TOPSIS = Technique for Order Preference by Similarity to Ideal Solution; COPRAS = Complex Proportional Assessment. The workflow documents normalisation, weighting, ranking, and robustness testing. Source: Author's own elaboration.

Table 19. Final Analytical Outputs and Interpretation Rules

Final output	What it reports	Interpretation rule
IAS score	0-100 composite score for each alternative	Higher means more attractive under stated assumptions
TOPSIS rank	Relative closeness to ideal solution	Used as primary ranking output
COPRAS rank	Relative utility based on benefit and cost criteria	Used as robustness comparison
Robustness class	Stable / partially stable / fragile	Based on rank changes under sensitivity tests
Data-confidence flag	High / medium / low	Based on the Data Quality Score

Note. IAS = Investment Attractiveness Score; TOPSIS = Technique for Order Preference by Similarity to Ideal Solution; COPRAS = Complex Proportional Assessment. Source: Author's own elaboration.

11. Expected Results

This section distinguishes between expected outcomes (the concrete outputs produced during the research), anticipated analytical patterns (the results the model is likely to reveal), potential impact (who can use the findings and how), and the value of negative or unexpected results. The wording is deliberately cautious: the study does not promise city-wide transformation or guaranteed investment returns. It promises a transparent and reproducible evaluation framework that can be tested on selected Casablanca sites and recalibrated for other contexts.

11.1 Expected Outcomes

The research is expected to produce five tangible outputs. Each output is tied to a specific research objective and can be inspected by the evaluator rather than accepted as a vague promise of improved knowledge.

Table 20. Expected Research Outcomes and Evaluation Criteria

Expected outcome	Concrete deliverable	Evaluation criterion
Risk-adjusted decision-support framework	A documented framework linking Mobility Potential, Financial Viability, Risk Exposure, Governance Quality, and Data Quality into the Investment Attractiveness Score (IAS).	The model variables, formulae, weights, assumptions, and scoring rules are explicit enough for another researcher to reproduce the analysis.
Pilot site-level dataset	A structured dataset for 5-10 candidate Casablanca sites, including spatial, mobility, financial, governance, risk, and data-quality indicators.	Each site includes a source inventory, inclusion/exclusion justification, data-quality score, and assumption register.
Location ranking matrix	A ranked comparison of selected project alternatives using IAS and MCDA ranking logic.	The ranking distinguishes raw mobility exposure from actual investment attractiveness.
Profitability-risk classification	A two-dimensional matrix classifying each site by financial viability and risk exposure.	The matrix identifies priority, fragile, low-return, and avoid/redesign cases.
Sensitivity and robustness outputs	Scenario tables showing how IAS and rankings change under alternative revenue, CAPEX, OPEX, traffic, discount-rate, governance, and data-quality assumptions.	The model reports unstable rankings as fragile rather than presenting them as definitive recommendations.

Note. The table distinguishes outputs that the study will produce from broader impacts that may follow after application. Source: Author's own elaboration.

11.2 Anticipated Analytical Patterns

The expected results are not arbitrary predictions. They follow logically from the literature and from the structure of the proposed model. The European Court of Auditors identifies fragmentation as a barrier to wider smart-city adoption, the OECD emphasises data-governance constraints in smart cities, and the World Bank PPP framework treats risk allocation and value-for-money assessment as central to project feasibility. These sources support the expectation that investment attractiveness will not be explained by traffic exposure alone (European Court of Auditors, 2023; OECD, 2023; World Bank, 2024b, 2024c).

- The highest-traffic sites are not expected to automatically receive the highest IAS. A high-exposure location may lose attractiveness once CAPEX, concession costs, data limitations, or regulatory friction are included.
- Revenue-rate assumptions, exposure coefficients, CAPEX, and discount-rate assumptions are expected to be the most sensitive financial drivers because they directly affect NPV, ROI, and payback period.
- Governance quality and data quality are expected to change site rankings in cases where mobility exposure is strong but legal feasibility, data reliability, or implementation predictability is weak.
- The sensitivity analysis is expected to reveal that some locations are robust across scenarios, while others are attractive only under optimistic revenue or low-cost assumptions.

These anticipated patterns directly answer the central 'so what' of the study: the framework should show when a site is merely visible and when it is genuinely investable.

11.3 Potential Impact

The immediate impact of the research is methodological: it converts a fragmented decision problem into a transparent evaluation workflow. The longer-term impact is practical: municipalities, PPP operators, and investors could use the framework as a pre-feasibility screening tool before committing to costly technical, legal, or financial due diligence. This matters because European municipalities report persistent infrastructure investment gaps, and smart-city projects face adoption barriers when indicators and coordination mechanisms remain fragmented (European Investment Bank, 2023; European Court of Auditors, 2023).

Table 21. Expected Beneficiaries and Five-Year Use Value

Beneficiary	How the output can be used	Five-year value if validated
Municipal authorities	Screen unsolicited PPP proposals, compare candidate sites, and identify data gaps before procurement.	More disciplined public-space monetisation and fewer investment decisions based only on visibility or political preference.
Private investors and infrastructure operators	Pre-screen locations using a risk-adjusted model before due diligence.	Lower wasted due-diligence cost and better separation between attractive sites and fragile sites.
Urban planners	Connect spatial mobility analysis with financial and governance feasibility.	A stronger bridge between transport planning, public-space management, and infrastructure finance.
Policy makers and regulators	Identify which governance or data constraints reduce investment attractiveness.	Evidence for targeted reforms in procurement clarity, data access, and PPP risk allocation.
Researchers	Replicate, refine, or challenge the IAS model using other cities or asset types.	A reusable analytical template for studying mobility-dependent infrastructure investment in comparable contexts.

Note. Five-year value indicates plausible application pathways, not guaranteed policy adoption. Source: Author's own elaboration.

11.4 Negative and Unexpected Results Plan

The study remains valuable even if the empirical results do not confirm the expected ranking logic. A rigorous proposal must anticipate negative findings rather than treating them as failure.

Table 22. Scientific Value of Negative or Unexpected Results

Possible result	Interpretation	Scientific value
IAS rankings are highly unstable across scenarios	The investment case is assumption-sensitive and cannot support deterministic recommendations.	Shows that smart infrastructure appraisal requires scenario-based decision-making rather than single-point scoring.
Mobility exposure explains little of final attractiveness	Cost structure, revenue conversion, governance, or data constraints dominate visibility.	Prevents the false assumption that traffic automatically equals investment value.
Governance scoring materially changes rankings	Institutional feasibility is not a background condition; it directly affects investability.	Strengthens the contribution to PPP and urban governance research.
Revenue data are insufficient for precise valuation	The model must rely on conservative, baseline, and optimistic scenarios.	Clarifies the empirical data gap and identifies what municipalities or operators should disclose for better appraisal.

Note. The study treats instability, weak data, or low revenue conversion as scientific findings rather than failures. Source: Author's own elaboration.

11.5 Validity, Generalisation, and Replicability of Results

The expected results will be interpreted within strict validity boundaries. Internal validity will be strengthened by explicit variable definitions, source hierarchy, normalisation rules, assumption registers, and sensitivity analysis. External validity will remain limited because the pilot sample focuses on selected Casablanca sites. The study will therefore avoid claiming that its rankings generalise to all Casablanca locations or to all Euro-Mediterranean cities.

Generalisation is conceptual rather than statistical. The specific numerical rankings will apply only to the selected sites and assumptions. The modelling logic, however, can be transferred to other mobility-dependent infrastructure contexts if local data, revenue assumptions, governance rules, and data-quality conditions are recalibrated. Replicability will be supported through a reproducibility package including the site list, variable dictionary, source inventory, data-quality register, scoring rubric, MCDA sheet, and sensitivity-analysis outputs.

11.6 Revised Expected Results Statement

The expected result of this research is not a universal investment formula. It is a bounded, transparent, and sensitivity-tested decision-support model that shows how mobility-derived demand, financial viability, governance quality, risk exposure, and data reliability jointly shape the investment attractiveness of selected smart urban infrastructure projects. If successful, the study will produce a concrete IAS framework, a pilot dataset, a location ranking matrix, a profitability-risk classification, and a reproducibility package. If results are unstable or counterintuitive, that outcome will still be analytically valuable because it will reveal which assumptions and data gaps prevent confident investment appraisal.

12. Academic and Practical Contributions

The contributions of this research are deliberately framed as bounded and evidence-based. The study does not claim to solve the full smart-city financing gap or to produce a universal investment formula. Its contribution is more precise: it shows how mobility-derived demand, financial viability, governance quality, regulatory risk, and data reliability can be operationalised within one transparent investment evaluation framework for mobility-dependent smart urban infrastructure.

This contribution responds directly to the research gap identified earlier: the relevant literatures provide strong individual tools, but weak integration. Smart-city research explains digital urban transformation, infrastructure finance provides project appraisal techniques, PPP literature clarifies risk allocation, and data-governance research explains the reliability and legality of urban data. However, these streams rarely converge into a site-level, risk-adjusted model that can compare actual infrastructure alternatives under stated assumptions.

12.1 Academic and Theoretical Contributions

First, the study contributes to applied urban economics by treating mobility exposure as an economic input rather than only a transport-planning variable. The research clarifies how traffic intensity, pedestrian flow, dwell time, visibility, and accessibility can be transformed into a Mobility Potential Index and then linked to financial viability. This advances the literature beyond descriptive mobility analysis by asking when spatial exposure becomes economically investable.

Second, the study contributes to infrastructure finance by extending CAPEX/OPEX, NPV, ROI, and payback logic to smaller-scale smart urban assets whose revenue depends on exposure, data quality, and governance conditions. This is important because many smart-city assets do not operate like conventional infrastructure concessions with stable tariff structures; their viability is more sensitive to advertising demand, concession terms, revenue conversion, and local implementation risk.

Third, the study makes a methodological contribution by developing a composite Investment Attractiveness Score rather than relying on a single profitability indicator. The model integrates five components: Mobility Potential, Financial Viability, Risk Exposure, Governance Quality, and Data Quality. This creates a replicable scoring logic that can be audited, recalibrated, and sensitivity-tested.

Fourth, the study contributes to smart-city governance research by making data quality and governance quality explicit determinants of investment evaluation. The OECD emphasises that smart-city success depends on data availability, effective data use, integrity, privacy, quality, safety, and interoperability; this research translates that governance logic into the mechanics of investment appraisal.

Fifth, the study contributes to PPP and public-investment evaluation by treating risk allocation, procurement transparency, permitting predictability, and enforcement reliability as investment-relevant variables. This is aligned with the World Bank view that PPP appraisal depends on value for money, affordability, market assessment, and risk allocation rather than financial return alone.

Finally, the study contributes to Euro-Mediterranean urban research by using Casablanca as an emerging-city empirical anchor and EU/Italian regulatory conditions as a benchmark. The contribution is not a broad cross-country comparison; it is a focused framework for examining how regulatory maturity and data-governance conditions alter the investability of similar mobility-dependent assets.

12.2 Practical and Managerial Contributions

The practical contribution is a pre-feasibility decision-support structure for actors who must compare competing smart-infrastructure opportunities before committing to procurement, due diligence, or deployment. The framework does not replace legal, financial, or engineering due diligence. It improves the quality of the first screening stage by making assumptions visible and comparable.

- **Municipal authorities:** can use the framework to prioritise candidate sites, evaluate unsolicited PPP proposals, and identify whether a project is weakened by data gaps, revenue uncertainty, or governance risk.
- **Private investors and infrastructure operators:** can use the IAS as a screening tool before committing resources to commercial negotiations, technical surveys, or legal review.
- **PPP and procurement units:** can use the governance and risk dimensions to identify where contract design, risk-sharing clauses, data-use permissions, or permitting procedures need clarification.
- **Urban planners:** can use the Mobility Potential and Data Quality indicators to connect infrastructure siting decisions with evidence on flows, accessibility, and spatial exposure.
- **Researchers:** can replicate, extend, or critique the model by applying the same variable dictionary, scoring rubric, and sensitivity-testing protocol to other cities or asset types.

12.3 Contribution-to-Gap Alignment

The table below shows how the contributions close the specific gap identified in the literature review rather than adding generic claims of relevance.

Table 23. Contribution-to-Gap Alignment

Identified gap	Contribution of this study	Concrete output	Primary users
Mobility studies estimate flows but rarely convert exposure into investment appraisal.	Links traffic, pedestrian flow, dwell time, visibility, and accessibility to financial viability.	Mobility Potential Index and exposure-to-revenue logic.	Urban planners; infrastructure operators; researchers.
Infrastructure finance uses NPV/ROI but often treats demand and governance as external assumptions.	Integrates CAPEX/OPEX, NPV, ROI, payback, revenue assumptions, and risk adjustment into one framework.	Financial Viability Score and sensitivity-tested cash-flow scenarios.	Investors; PPP operators; finance analysts.
Smart-city research often emphasises technology and governance but under-specifies financial viability.	Positions financial viability as a first-order condition of smart-infrastructure evaluation.	IAS framework linking technical attractiveness to investment feasibility.	Smart-city researchers; municipalities.
PPP/governance research explains risk allocation but rarely connects it to mobility-derived revenue assumptions.	Operationalises procurement transparency, risk allocation, permitting, and enforcement as governance variables.	Governance Quality Score and Risk Exposure Score.	Procurement units; policymakers; PPP specialists.
Data-governance literature identifies quality and compliance challenges but rarely turns them into	Treats data quality as a confidence condition affecting the reliability of investment rankings.	Data Quality Score and reproducibility package.	Data-governance teams; researchers; municipal analysts.

Identified gap	Contribution of this study	Concrete output	Primary users
valuation variables.			

Note. The table prevents contribution claims from becoming generic by tying each contribution to a specific gap. Source: Author's own elaboration.

12.4 Scope and Realistic Boundary of the Contribution

The contribution is intentionally bounded. The study does not claim to produce a universal model for all smart-city assets, nor does it claim to replace political judgement, legal due diligence, engineering feasibility studies, or investor negotiation. Its value lies in making the pre-feasibility stage more disciplined. It helps decision-makers distinguish between infrastructure that is merely visible, technologically attractive, or politically preferred, and infrastructure that is financially viable, institutionally feasible, data-supported, and robust under uncertainty.

This boundary strengthens rather than weakens the proposal. By focusing on mobility-dependent assets and pilot sites in Casablanca, the research remains empirically manageable. By using EU/Italian regulatory conditions as a benchmark rather than a second full empirical case, it preserves comparative relevance without overextending the design. The result is a realistic contribution: a tested analytical architecture that future studies can scale, adapt, or challenge with richer data.

13. Limitations and Risk Mitigation

This study is deliberately designed as a bounded proof-of-concept rather than a city-wide investment appraisal or a full legal due-diligence exercise. The limitations below do not invalidate the project; they define the conditions under which its findings should be interpreted. Following standard research-design logic, the section distinguishes delimitations chosen by the researcher from limitations imposed by data access, measurement uncertainty, time, budget, and methodological constraints (Theofanidis & Fountouki, 2018).

13.1 Scope and Delimitation of the Study

The first limitation is conceptual scope. The study does not evaluate all smart city technologies, all public infrastructure assets, or the total social value of urban digitalisation. It focuses only on mobility-dependent smart urban infrastructure, such as digital out-of-home advertising panels, smart bus shelters, connected street furniture, and mobility-linked urban nodes. This delimitation is necessary because the proposed analytical model requires a clear link between spatial exposure, revenue assumptions, CAPEX, OPEX, governance conditions, and data quality.

The consequence is that the conclusions will not be generalisable to smart city systems whose value is not exposure-dependent, such as water-management platforms, energy grids, public-safety sensors, or purely administrative digital services. The mitigation strategy is to frame the research output as a specialised investment-evaluation framework for a specific asset class, not as a universal smart city evaluation model.

13.2 Data Availability and Revenue Estimation Constraints

The most important practical limitation concerns access to primary revenue data. Street-level advertising yields, concession payments, occupancy rates, and operator-level commercial performance are often proprietary. If such data are unavailable, projected revenue will depend on documented assumptions, benchmarks, or conservative scenarios rather than direct operator records. This is the highest-risk limitation because revenue conversion is the bridge between mobility exposure and investment attractiveness.

The study mitigates this risk in three ways. First, the model separates Mobility Potential from Financial Viability, so traffic exposure is never treated as revenue by itself. Second, all revenue assumptions will be placed in an assumption register and tested under conservative, baseline, and optimistic scenarios. Third, the final interpretation will report whether a site remains attractive across scenarios or becomes attractive only under optimistic monetisation assumptions.

13.3 Measurement Error in Mobility and Exposure Data

The second methodological limitation concerns measurement error in mobility and exposure indicators. Traffic intensity, pedestrian flow, dwell time, directionality, and line-of-sight exposure are difficult to measure perfectly without continuous sensor data. When official traffic counts are unavailable, the study may rely on OpenStreetMap-derived spatial proxies, short field observations, or SUMO simulation outputs. These substitutes are useful, but they are less precise than continuous observed mobility data.

This limitation is mitigated through source labelling and triangulation. The dataset will explicitly distinguish observed, simulated, estimated, and assumption-based values. Each site will receive a Data Quality Score based on completeness, granularity, recency, consistency, legal usability, and source reliability. The OECD's smart city data governance work reinforces the need to check the quality and limitations of data and to evaluate wider policy implications before using data for urban decisions (OECD, 2023).

13.4 Sampling and Generalisability Limits

The study uses 5 to 10 purposively selected candidate sites in Casablanca. This is sufficient for pilot-testing the logic of the Investment Attractiveness Score, but it is not statistically representative of all Casablanca infrastructure locations. The sample is designed for analytical calibration, not for city-wide inference.

The mitigation strategy is to avoid overclaiming. The study will not conclude that one type of site is universally superior across Casablanca. It will conclude only whether the proposed framework can rank selected project alternatives transparently under stated assumptions. Future work can expand the sample, include additional districts, and test whether the same ranking logic holds across other Euro-Mediterranean cities.

13.5 Subjectivity in Governance and Regulatory Scoring

A further limitation concerns the conversion of legal and governance conditions into numerical scores. Procurement transparency, PPP risk allocation, permitting predictability, enforcement reliability, and data-protection constraints are complex institutional variables. Reducing them to a 0–5 scale risks oversimplification.

The mitigation strategy is to use a predefined documentary scoring rubric rather than ad hoc judgement. Each governance or regulatory score must be supported by a legal text, institutional document, procurement rule, or clearly stated assumption. The model will also separate Governance

Quality from Risk Exposure to avoid double-counting the same institutional weakness twice. This is consistent with PPP appraisal logic, where risk allocation affects value for money and should be explicitly assigned and interpreted rather than hidden inside a general feasibility judgement (World Bank, 2024c).

13.6 MCDA Weighting and Model-Dependence

The analytical model relies on weighted scoring and multi-criteria decision analysis. This introduces model-dependence: different weights, normalisation procedures, or ranking algorithms may produce different site rankings. AHP weights may also reflect expert judgement and can therefore contain subjective bias.

This limitation will be mitigated through sensitivity and robustness checks. The study will test alternative weight scenarios, vary uncertain inputs, and compare ranking stability where feasible. If TOPSIS and COPRAS produce materially different rankings, this will be reported as a methodological finding rather than hidden. The final IAS will therefore be interpreted as a transparent decision-support score, not as an objective or deterministic truth.

13.7 Time, Budget, and Access Constraints

The research is constrained by the time and resource limits of a Master's-level project. It cannot conduct a full longitudinal study, continuous sensor-based traffic monitoring, city-wide field observation, operator-level commercial audits, or full legal due diligence across multiple countries. These activities would require institutional access, larger funding, and a longer research horizon.

The mitigation strategy is methodological discipline. The project uses a bounded pilot design, transparent assumptions, reproducible scoring rules, and sensitivity analysis. These choices make the study feasible without pretending to deliver a complete investment appraisal for every smart infrastructure opportunity in Casablanca or the Euro-Mediterranean region.

13.8 Impact of Limitations on the Conclusions

The limitations affect the strength of the conclusions in three ways. First, the findings will be valid primarily for the selected pilot sites and stated assumptions. Second, the model will support comparative ranking and scenario-based interpretation, not causal claims about all smart infrastructure investments. Third, the framework will be transferable as an analytical architecture, but its numerical results must be recalibrated when applied to other cities, asset types, cost structures, or regulatory environments.

These limits are not fatal. They make the contribution more precise. The study's value lies in showing how fragmented evidence—mobility exposure, financial viability, risk, governance, and data quality—can be organised into a transparent, auditable and sensitivity-tested investment-evaluation framework.

Table 24. Limitations, Validity Threats, and Mitigation Strategies

Limitation	Threat to validity	Mitigation strategy
Limited site sample	Weak external validity beyond selected Casablanca locations	Treat the study as proof-of-concept; document site-selection criteria; avoid city-wide claims.

Limitation	Threat to validity	Mitigation strategy
Revenue data constraints	Risk of confusing exposure with monetisable demand	Use assumption register; conservative/baseline/optimistic scenarios; sensitivity testing.
Mobility-data uncertainty	Measurement error in traffic, pedestrian flow, dwell time, and exposure	Triangulate observed, simulated, and proxy data; assign Data Quality Scores.
Governance scoring subjectivity	Risk of arbitrary legal-to-numeric conversion	Use documentary scoring rubric; justify each score; separate governance from risk exposure.
MCDA weight sensitivity	Rankings may change under alternative weights or methods	Run weight sensitivity; compare ranking stability; report fragile rankings.

Note. Limitations are treated as bounded risks to interpretation, not as flaws that invalidate the project. Source: Author's own elaboration.

14. Ethical and Governance Considerations

This section sets out the ethical and governance safeguards for the proposed study. The project is expected to be low risk because it relies primarily on secondary, aggregated, simulated, and documentary data. However, if expert judgements are collected for AHP weighting, scoring validation, or methodological review, those experts will be treated as human participants. The study will therefore apply a formal consent, confidentiality, data-protection, and institutional review protocol before any participant-related data are collected.

14.1 Ethical Framework

The study will be guided by the four principles of autonomy, non-maleficence, beneficence, and justice. Autonomy requires voluntary participation and informed decision-making. Non-maleficence requires the researcher to avoid foreseeable harm, including reputational, professional, or confidentiality harm. Beneficence requires the study to generate proportionate academic and practical value. Justice requires fair recruitment and non-discriminatory treatment of participants. These principles are consistent with the Belmont Report's emphasis on respect for persons, beneficence, and justice in research involving human subjects, and with the broader four-principles approach associated with Beauchamp and Childress (Beauchamp & Childress, 2019; U.S. Department of Health and Human Services, 1979).

14.2 Institutional Ethics Approval

Before any expert interview, AHP questionnaire, professional judgement, or validation exercise is conducted, the full research protocol will be submitted to the relevant university ethics committee or Institutional Review Board. The submission will include the participant information sheet, consent form, recruitment script, expert questionnaire or interview guide, data-management plan, risk assessment, and confidentiality protocol. Research involving human participants should be reviewed independently before data collection; the WHO Research Ethics Review Committee, for example, reviews human-participant research to ensure that supported studies meet ethical standards (World Health Organization, n.d.).

If the university determines that the study is exempt because it uses only secondary, aggregated, or simulated data, the researcher will request written confirmation of exemption or ethics clearance. The study will not self-declare ethical compliance without institutional confirmation.

14.3 Informed Consent Procedure

If expert participants are involved, informed consent will be obtained before data collection. Participation will be voluntary and free from coercion, professional pressure, or implied obligation. Each participant will receive a written information sheet explaining:

- the purpose of the research and why the participant was invited;
- the expected duration and format of participation;
- the type of information requested and how it will be used;
- the voluntary nature of participation and the right to decline any question;
- the right to withdraw before anonymisation or aggregation;
- the confidentiality safeguards and limits of confidentiality;
- the storage, retention, and destruction procedures;
- the researcher and supervisor contact details.

No participant will be asked to disclose confidential company information, trade secrets, private contract terms, non-public municipal records, personal client information, or identifiable mobility data. If a participant begins to disclose restricted or commercially sensitive information, the researcher will stop the discussion, mark the material as excluded, and remove it from the dataset.

14.4 Participant Selection, Vulnerability, and Justice

The study will recruit only adult professionals or academic experts if expert input is required. Eligible participants may include urban planners, infrastructure-finance professionals, PPP or procurement specialists, smart-city practitioners, data-governance specialists, or academic researchers. Participants will be selected because of relevant expertise, not because of personal characteristics such as gender, nationality, ethnicity, religion, or socioeconomic background.

Children, minors, vulnerable adults, patients, persons unable to provide informed consent, and individuals in dependent or coercive relationships with the researcher will be excluded. This exclusion is justified because the study does not require vulnerable populations to answer the research question. Where feasible, the expert panel will seek professional diversity across planning, finance, governance, and data domains. The study will not claim demographic representativeness because expert input will support methodological weighting or validation, not population inference.

14.5 Confidentiality, Anonymisation, and Commercial Sensitivity

Expert responses will be treated as confidential. Participants will be assigned pseudonymous codes such as E01, E02, and E03. Names, email addresses, institutional identifiers, signed consent forms, and response files will be stored separately. The final thesis or proposal will not identify participants by name unless explicit written permission is obtained.

The study will distinguish carefully between anonymisation and pseudonymisation. Anonymised data must make re-identification impossible in practice; pseudonymised data remain protected because identities may be recoverable if the key exists. European Commission guidance states that personal data rendered anonymous are no longer personal data only when the individual is no longer identifiable and anonymisation is irreversible. Because the expert sample may be small, the study will avoid overly precise role descriptions that could indirectly identify participants (European Commission, n.d.).

Commercial confidentiality is a specific governance risk in this project. Experts in PPP, DOOH, advertising, procurement, or municipal infrastructure may possess non-public information. The protocol will therefore explicitly instruct participants not to disclose confidential pricing, concession terms, revenue figures, contract clauses, tender information, or internal strategy. Any accidentally disclosed confidential information will be excluded from analysis and documented in an audit trail.

14.6 Data Protection and Legal Compliance

The study will comply with applicable data-protection principles. For the Moroccan context, the research will respect the logic of Loi 09-08 and the role of the CNDP as the authority responsible for the protection of personal data. For EU-related data or expert participation, the study will follow GDPR-aligned principles, including lawfulness, fairness, transparency, data minimisation, purpose limitation, storage limitation, integrity, and confidentiality (Commission Nationale de Controle de la Protection des Donnees a Caractere Personnel, n.d.; European Union, 2016).

The study will not collect individual GPS traces, mobile-phone records, personal advertising identifiers, vehicle identification records, or identifiable movement histories. Mobility indicators will be aggregated, simulated, or non-identifiable. If a dataset cannot be used lawfully, ethically, and at a sufficiently aggregated level, it will be excluded. Article 32 GDPR requires security measures appropriate to risk, including pseudonymisation and encryption where relevant; these principles will guide the project data-security protocol (European Union, 2016).

14.7 Data Storage, Access Control, and Retention

Research data will be stored in a structured and access-controlled project folder. The storage architecture will separate raw secondary datasets, cleaned datasets, expert responses, consent forms, coding sheets, financial assumption registers, MCDA calculation files, and sensitivity-analysis outputs. Identifiable participant information will be stored separately from analytical files.

Files containing consent forms, contact information, or non-anonymised expert responses will be encrypted and accessible only to the researcher and, where required, the academic supervisor. The project will avoid unsecured cloud storage unless approved by the university. Data will be retained only for the period required by university policy or academic verification. After the retention period, identifiable files will be securely deleted. Aggregated, anonymised, or non-identifiable analytical outputs may be retained for reproducibility if permitted by the ethics approval.

14.8 Risk-Benefit and Proportionality Assessment

The expected risk level is low, but not zero. The main risks are professional confidentiality risk, indirect re-identification risk, data-security risk, analytical misuse risk, and expert-bias risk. These risks are proportionate because the study is non-invasive, does not involve vulnerable populations, does not collect individual mobility traces, and produces a practical framework for more transparent infrastructure-investment evaluation.

Table 25. Ethical Risks, Potential Harm, and Mitigation Strategies

Risk	Potential harm	Mitigation strategy
Professional confidentiality risk	Experts may disclose non-public commercial, PPP, or municipal information.	Warn participants before data collection; stop disclosure; exclude confidential material; document exclusion in the audit trail.
Indirect re-identification risk	Small expert samples may make participants identifiable by role or institution.	Use pseudonyms, broad professional labels, and aggregated reporting.
Data-security risk	Unauthorised access to expert responses or assumptions.	Encrypt identifiable files, separate consent forms from responses, and restrict access.
Analytical misuse risk	IAS may be misread as a definitive investment decision.	Present IAS as a pre-feasibility decision-support framework, not a substitute for legal or financial due diligence.
Expert-bias risk	AHP weights may reflect professional background or sectoral bias.	Diversify experts where feasible and test alternative weighting scenarios through sensitivity analysis.

Note. The risk assessment applies mainly if expert judgement is collected; aggregated and simulated mobility data remain lower risk. Source: Author's own elaboration.

14.9 Governance of the Analytical Model

The ethical governance of this project also concerns the analytical model itself. The Investment Attractiveness Score will not be presented as a black-box or value-neutral algorithm. Every indicator, data source, assumption, score, weight, and sensitivity test will be documented in a reproducibility package. This includes the variable dictionary, source inventory, scoring rubric, financial assumption register, MCDA calculation sheet, and sensitivity-analysis outputs.

The model will include an explicit warning against overinterpretation. It will support pre-feasibility screening, but it will not replace legal due diligence, financial negotiation, stakeholder consultation, municipal planning judgement, or political accountability. This safeguard prevents the study from converting a structured academic model into an unjustified automated decision tool.

14.10 Conflict of Interest and Research Integrity

The researcher will disclose any prior academic or professional experience related to Casablanca, mobility analysis, advertising infrastructure, financial modelling, or smart-city investment evaluation. No private operator, supplier, municipality, or expert participant will be allowed to influence the scoring rubric, suppress negative findings, or control the interpretation of results.

All assumptions will be documented transparently. If data are missing, simulated, estimated, or assumption-based, the study will label them as such. This is essential to avoid false precision and to preserve the credibility of the proposed framework.

14.11 Incident Response Protocol

If an ethical or data-protection incident occurs, such as accidental disclosure of confidential information, unauthorised access to files, or participant withdrawal after data collection, the researcher will apply a documented incident-response procedure. The affected material will be isolated, excluded from analysis where required, and reported to the supervisor or ethics committee according to university policy. If the incident concerns personal data, the researcher will follow the applicable data-protection reporting requirements and will not continue using the affected material until the issue has been resolved.

15. Feasibility and Timeline

The research is feasible because it relies on established tools and methods: GIS/SUMO for mobility and spatial modelling, standard DCF indicators for financial evaluation, AHP/TOPSIS/COPRAS for scoring and ranking, and documentary analysis for regulatory comparison. The main feasibility risk is data access, especially for primary revenue data and detailed municipal traffic records.

Table 26. Feasibility Timeline by Research Phase

Phase	Tasks	Estimated duration
Phase 1 - Literature and framework finalisation	Refine literature review, define variables, finalise IAS architecture	4-6 weeks
Phase 2 - Data collection	Collect traffic, spatial, financial, regulatory, and macroeconomic data	6-8 weeks
Phase 3 - Model construction	Build DCF model, normalise indicators, prepare AHP weighting instruments	4-6 weeks
Phase 4 - Scoring and sensitivity analysis	Run IAS, TOPSIS/COPRAS ranking, and scenario tests	4-6 weeks
Phase 5 - Interpretation and writing	Develop findings, implications, limitations, and final manuscript	6-8 weeks

Note. Durations are estimates and may be adjusted after supervisor feedback, ethics review, and data-access confirmation. Source: Author's own elaboration.

16. Conclusion

Smart urban infrastructure investment increasingly forces cities and investors to decide at the intersection of mobility demand, financial viability, regulatory feasibility, governance quality, and data reliability. Yet these dimensions are still too often evaluated separately: traffic analysis identifies exposure, financial modelling estimates returns, legal review checks compliance, and governance assessment examines implementation risk. This proposal addresses that fragmentation by developing a risk-adjusted Investment Attractiveness Score for mobility-dependent smart urban infrastructure. Anchored in Casablanca and benchmarked against EU/Italian governance conditions, the study offers a bounded, reproducible, and empirically calibratable framework for converting mobility exposure into investment-relevant evidence.

The contribution of this research is not the claim that mobility data, financial modelling, PPP governance, or MCDA are new. Its contribution lies in integrating them into one operational decision logic. The proposed methodology shows how Mobility Potential, Financial Viability, Risk Exposure, Governance Quality, and Data Quality can be measured, weighted, stress-tested, and translated into a transparent ranking of project alternatives. This makes the study academically relevant for urban economics, infrastructure finance, smart-city governance, and MCDA research, while offering practical value for municipalities, investors, PPP operators, and urban planners. The final purpose is clear: to prevent smart infrastructure decisions from being driven by visibility, technological appeal, or isolated profitability estimates alone. This research turns urban exposure into an auditable investment argument, distinguishing infrastructure that is merely visible from infrastructure that is genuinely investable.

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